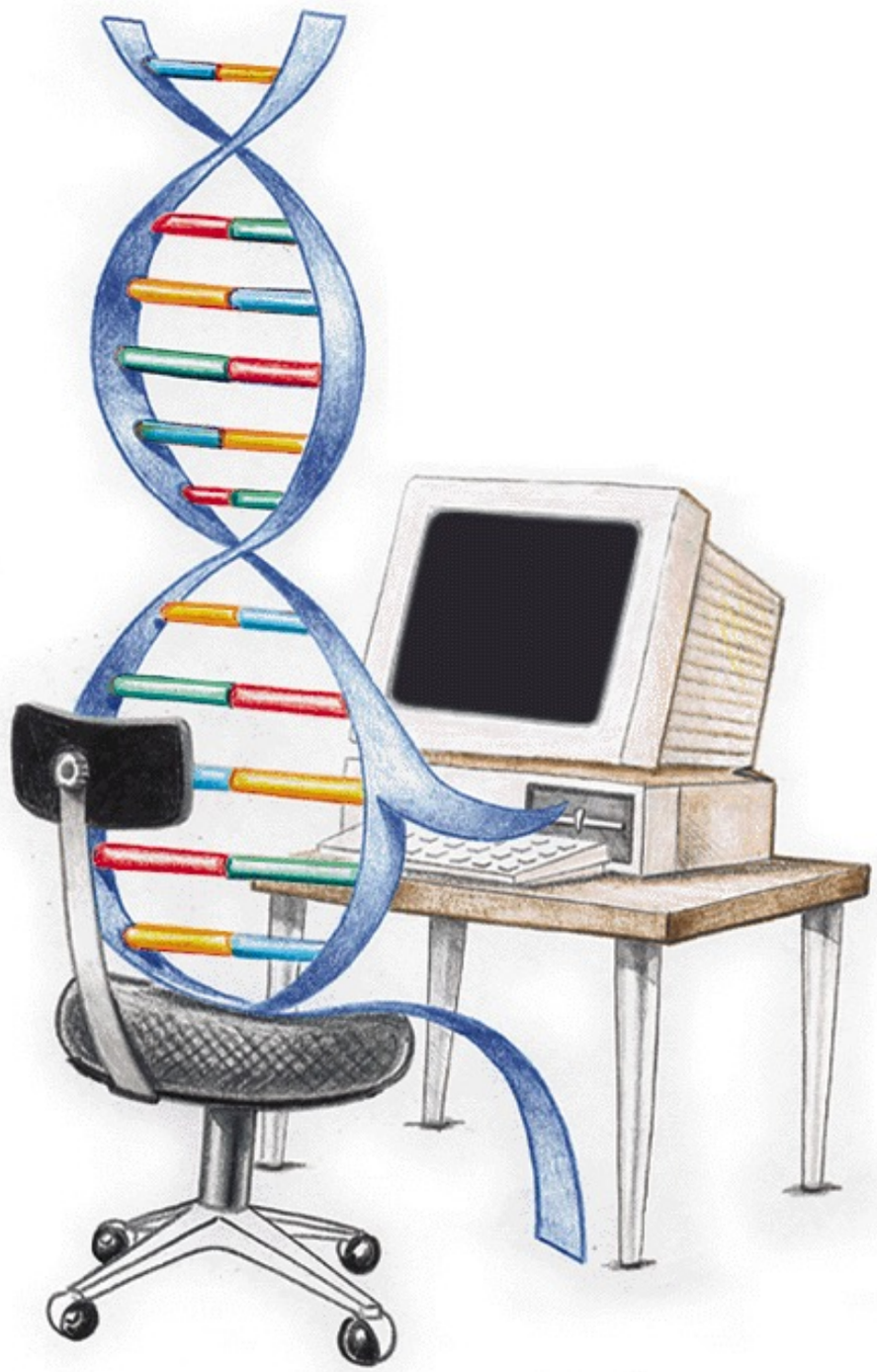
Genetic Algorithm



CSE-2530:
Computational Intelligence

Stephanie Tan

(Slides adapted from Pradeep K. Murukannaiah,
Joost Broekens, and Jazmin Salazar)

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Recap

After the previous lecture, you should be able to

- Explain the intuition behing **evolutionary computing**
- Recognize **problems** that can be modeled via evolutionary computing methods
- Enumerate the steps of a **genetic algorithm** (GA)
- Describe the key GA operations: **selection**, **cross-over**, and **mutation**
- Apply GA to **optimization problems**

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Learning Objectives

After today's lecture you should be able to:

- Analyze GA via **peformance graphs**
- Apply GA to a practical **scheduling problem**
- Recognize **multi-objective optimization** problems
- Describe **NSGA-II**, a GA for multi-objective optimization

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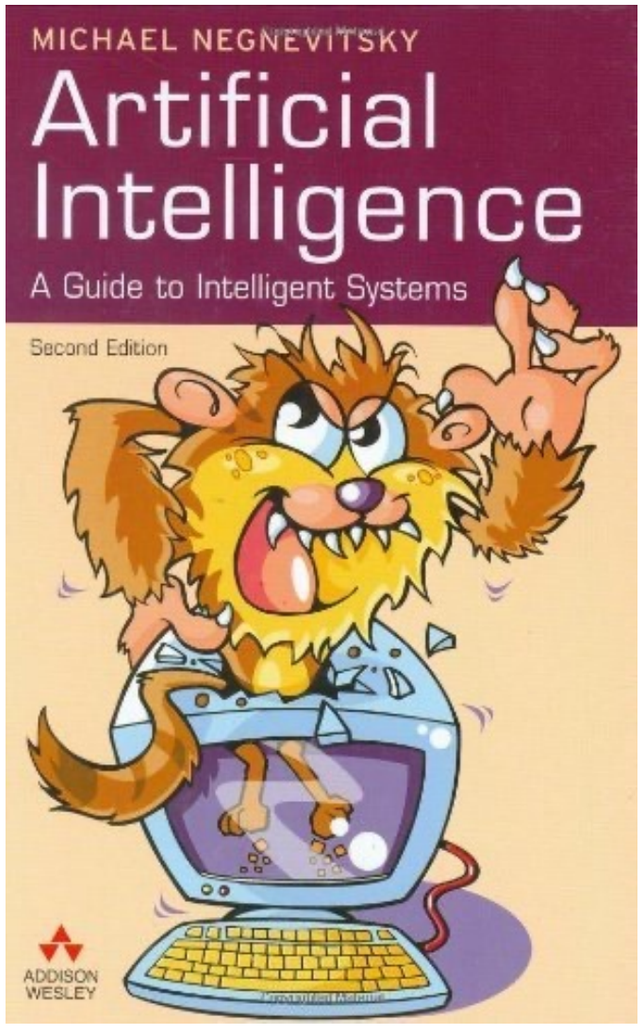
Study Material



Chapter 7: Sections 7.5

Lecture slides

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Performance Graphs

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Performance Graphs

- How do we know whether a solution we found is good?

- Performance graph:
  - Average fitness of the population over time
  - Fitness of the fittest individual per population

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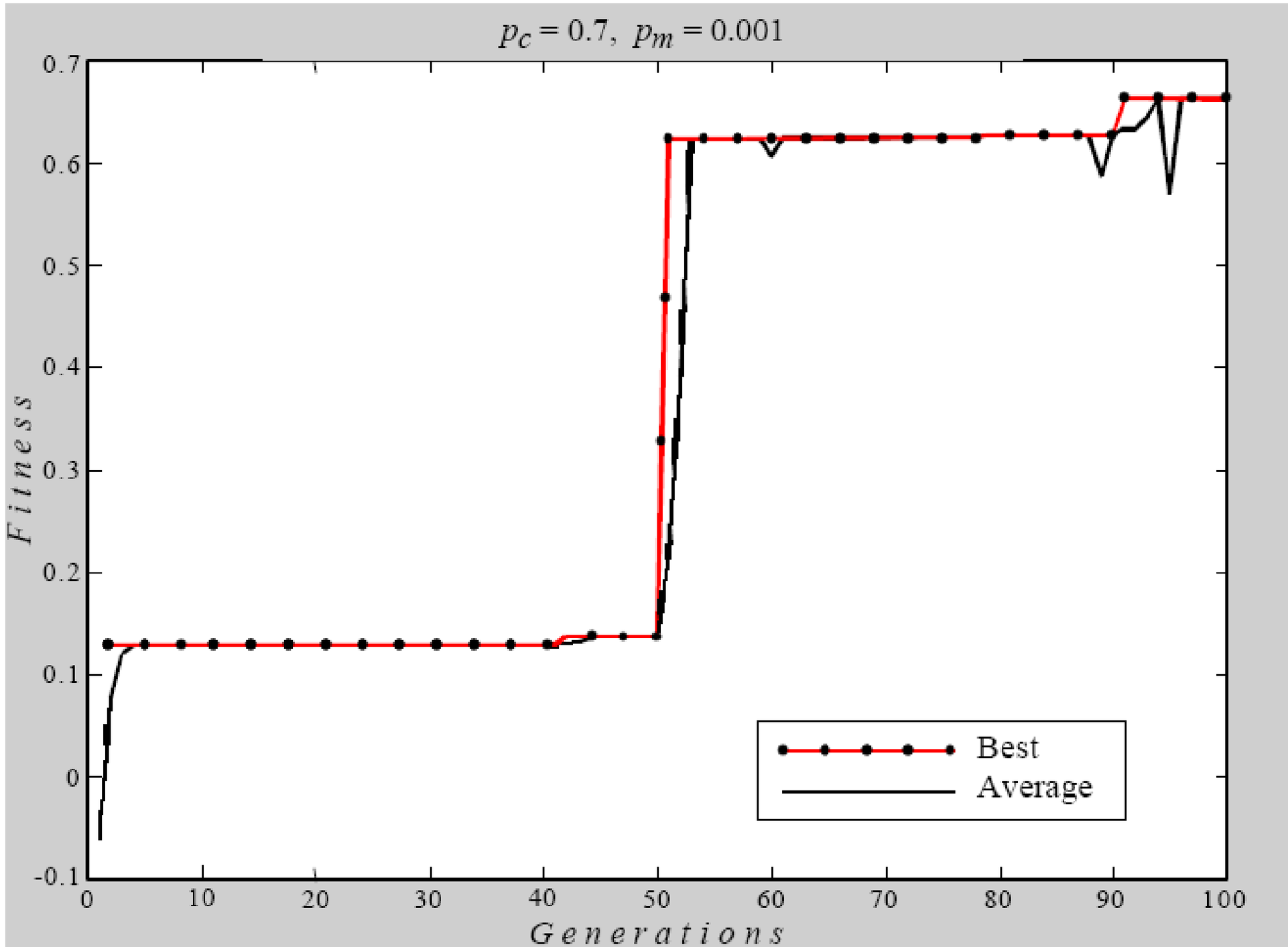
Performance Graphs

$P_c = 0.7$
$P_m = 0.001$



100 generations, 6 chromosomes, local maximum

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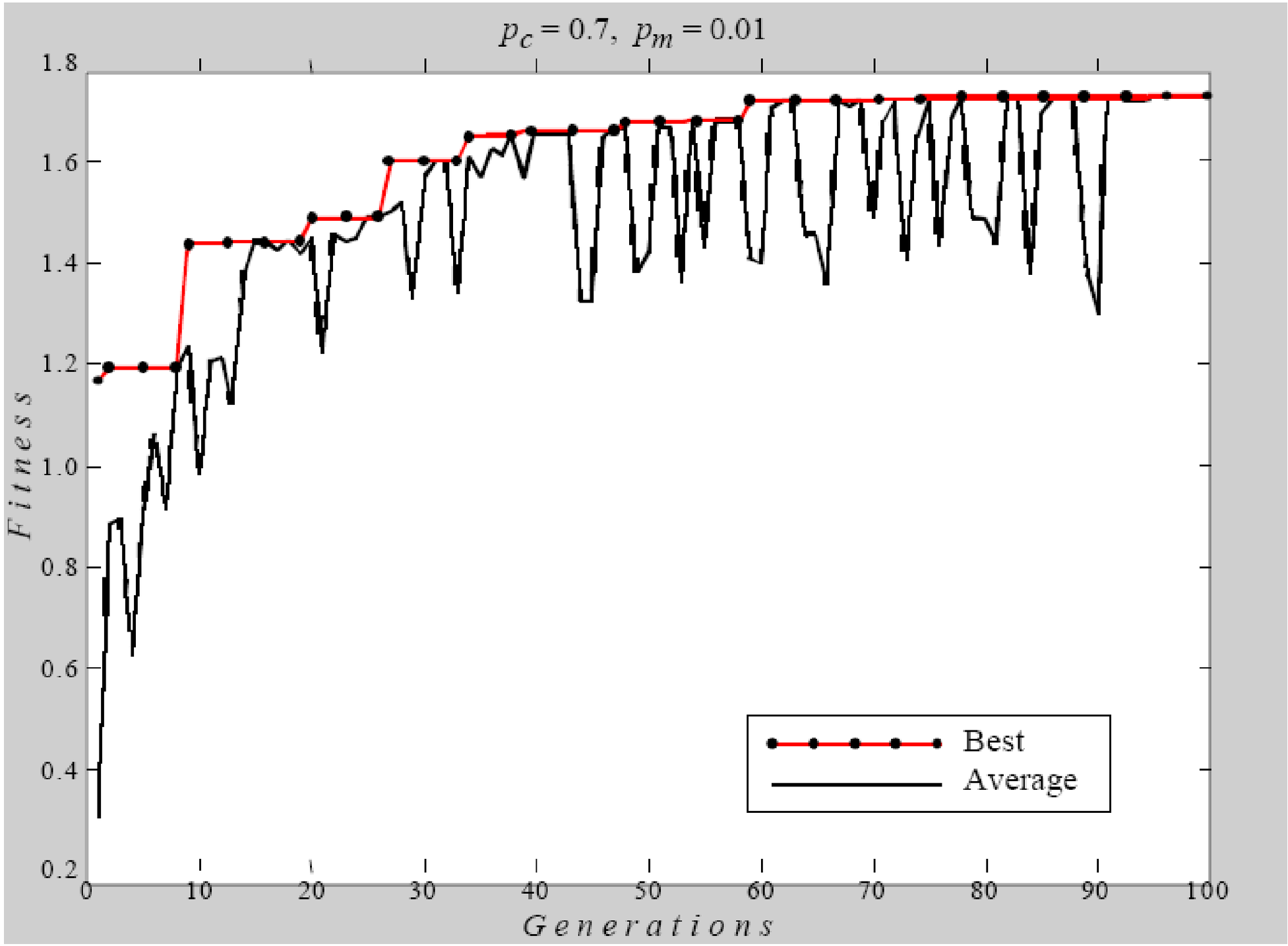
Performance Graphs

$P_c = 0.7$

$P_m = 0.01$



100 generations, 6 chromosomes, global maximum

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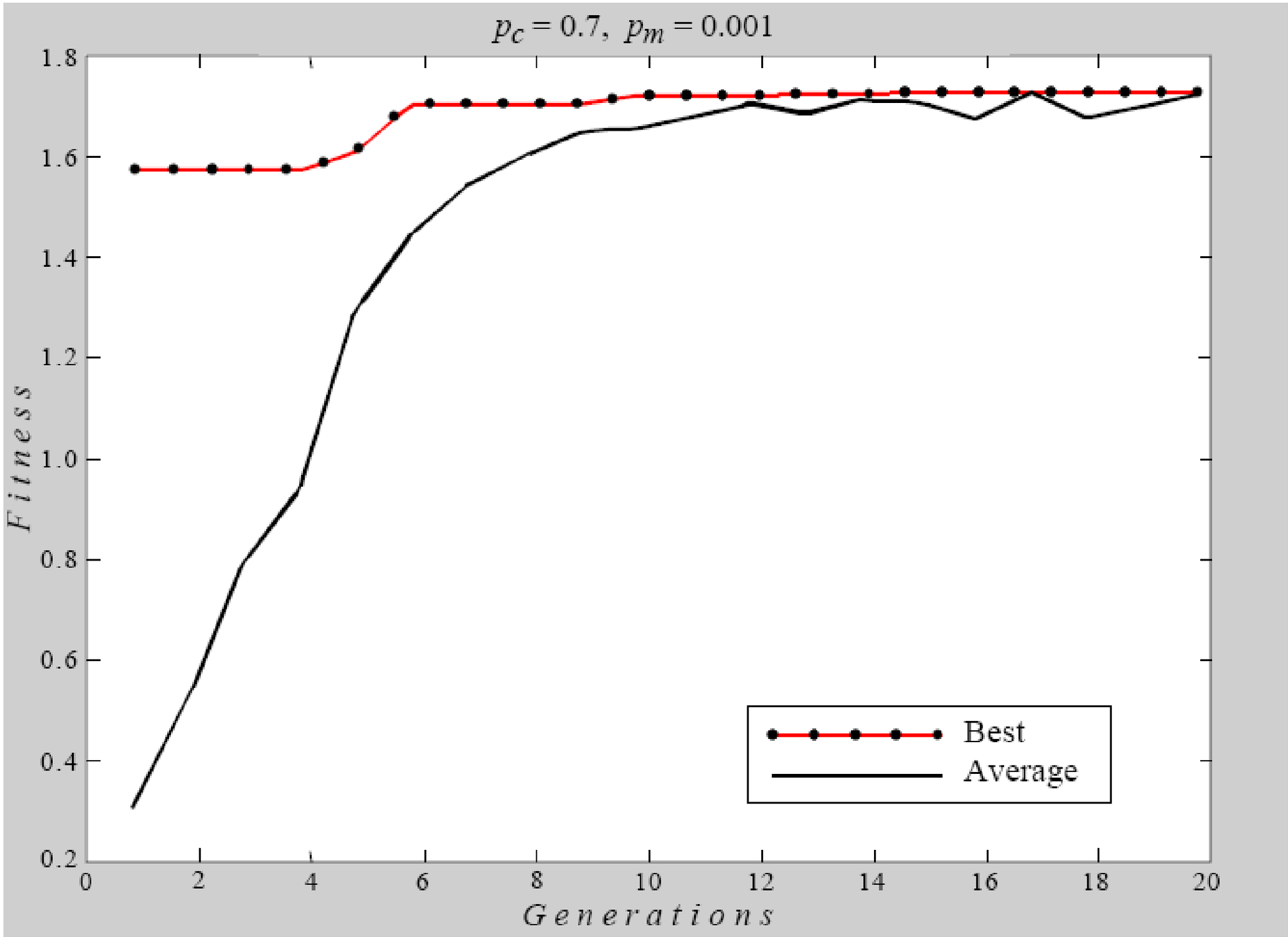
Performance Graphs

$P_c = 0.7$
$P_m = 0.001$



20 generations, 60 chromosomes, global maximum

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A Case Study

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GA Case Study: Maintenance Schedule for Power systems

- Scheduling is complex
  - A lot of requirements and constraints
  - NP complete (and not all solutions satisfy the requirements – remember the TUDelft schedule?)

- Scenario: **Yearly** maintenance of power units in a power system
  - Should be done to prevent unexpected breakdowns
  - A unit cannot generate power during its maintenance → shortage
  - Units should be maintained based on the capacity of the system to generate the required power, nevertheless

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Recap: GA steps

A typical process of the GA development includes the following steps:

- Step 1: Specify the problem, define constraints and optimum criteria.
- Step 2: Represent the problem domain as a chromosome.
- Step 3: Define a fitness function to evaluate the chromosome's performance.
- Step 4: Construct the genetic operators.
- Step 5: Run the GA and tune its parameters

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Problem

- Power system units operate continuously but require preventive maintenance
- Maintenance requires taking units offline, which reduces system security
- The goal of maintenance scheduling is to plan outages so overall system security is maximized
- System security is measured by the net reserve during each maintenance period
- Net reserve represents the safety margin between available generation and expected demand

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Problem

- Definition:

Net reserve = total installed capacity
- capacity of units under maintenance
- maximum forecasted load during that period

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Problem

- Definition:

Net reserve = total installed capacity
- capacity of units under maintenance
- maximum forecasted load during that period

- Example
- Total installed capacity = 150 MW
- Unit on maintenance = 20 MW
- Maximum forecasted load = 100 MW
- Net reserve = 150 – 20 – 100 = 30 MW

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Problem

- Definition:

Net reserve = total installed capacity
- capacity of units under maintenance
- maximum forecasted load during that period

- Example
• Total installed capacity = 150 MW
• Unit on maintenance = 20 MW
• Maximum forecasted load = 100 MW
• Net reserve = 150 – 20 – 100 = 30 MW

- Key implication
• Maintenance schedules must ensure net reserve stays sufficiently positive
• Low or negative net reserve increases blackout risk
• Optimization methods like genetic algorithms must model these constraints correctly

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Case Study: Maintenance Schedule

- Expected loads per quarter: 80 MW, 90 MW, 65 MW, 70 MW
-
- Total capacity: 150 MW
- Unit capacities:

Unit number	*Unit capacity, MW*	*Number of intervals required for unit maintenance*
1	20	2
2	15	2
3	35	1
4	40	1
5	15	1
6	15	1
7	10	1

- Maintenance takes 1 or 2 quarters (no interruption)

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Case Study: Maintenance Schedule

Problem:

Generate a maintenance schedule for the whole year

- How would you encode solutions to this problem?
- What fitness function would you use?
- How would you perform cross-over and mutation?

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Maintenance Schedule: Encoding

How does a candidate solution look like in this scenario?

- Candidate solution = Schedule
- Multiple units can be maintained at the same time

How do we encode the candidate solutions into chromosomes?

- Encode each unit into a 4 bit string
- Each bit indicates the status of the unit in the quarter
  - 1 = in maintenance for that quarter
  - 0 = otherwise
- Chromosome of 28 bits

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7
0 1 1 0	0 0 1 1	0 0 0 1	1 0 0 0	0 1 0 0	0 0 1 0	1 0 0 0

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Maintenance Schedule: Encoding

Redefine the syntax:
A gene is an **indivisible** 4-bit string

Gene pools:

Unit 1 is in maintenance for the first two quarters

Unit 1:	1 1 0 0	0 1 1 0	0 0 1 1	
Unit 2:	1 1 0 0	0 1 1 0	0 0 1 1	
Unit 3:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 4:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 5:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 6:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 7:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7
0 1 1 0	0 0 1 1	0 0 0 1	1 0 0 0	0 1 0 0	0 0 1 0	1 0 0 0

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Maintenance Schedule: Fitness

Fitness function

A solution is a schedule for the whole year, for all units

Even if some units are in maintenance, some **reserve** of power is desirable:
Net reserve >= 0

- Compute the reserve per quarter:
  Net reserve =
  = total capacity – maintained unit capacity – expected load
- Take the **minimum** value of Net reserve over the 4 quarters (worst case scenario)
- If Net reserve < 0 : fitness = 0

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Maintenance Schedule: Fitness

Unit number	*Unit capacity, MW*
1	20
2	15
3	35
4	40
5	15
6	15
7	10

Compute the reserve per quarter:

Net reserve =

= total capacity – maintained unit capacity – expected load

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7
0 1 1 0	0 0 1 1	0 0 0 1	1 0 0 0	0 1 0 0	0 0 1 0	1 0 0 0

Quarter 1: Net reserve = 150MW – U4 – U7 – 80MW = 20MW
Quarter 2: Net reserve = 150MW – U1 – U5 – 90MW = 25MW
Quarter 3: Net reserve = 150MW – U1 – U2 – U6 – 65MW = 35MW
Quarter 4: Net reserve = 150MW – U2 – U3 – 70MW = 30MW

→ Fitness = 20 (minimum across the quarters)

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Maintenance Schedule: Crossover

Crossover operator does not cut across genes

Parent 1						
0 1 1 0	0 0 1 1	0 0 0 1	1 0 0 0	0 1 0 0	0 0 1 0	1 0 0 0
Parent 2						
1 1 0 0	0 1 1 0	0 1 0 0	0 0 0 1	0 0 1 0	1 0 0 0	0 1 0 0
Child 1						
0 1 1 0	0 0 1 1	0 0 0 1	1 0 0 0	0 0 1 0	1 0 0 0	0 1 0 0
Child 2						
1 1 0 0	0 1 1 0	0 1 0 0	0 0 0 1	0 1 0 0	0 0 1 0	1 0 0 0

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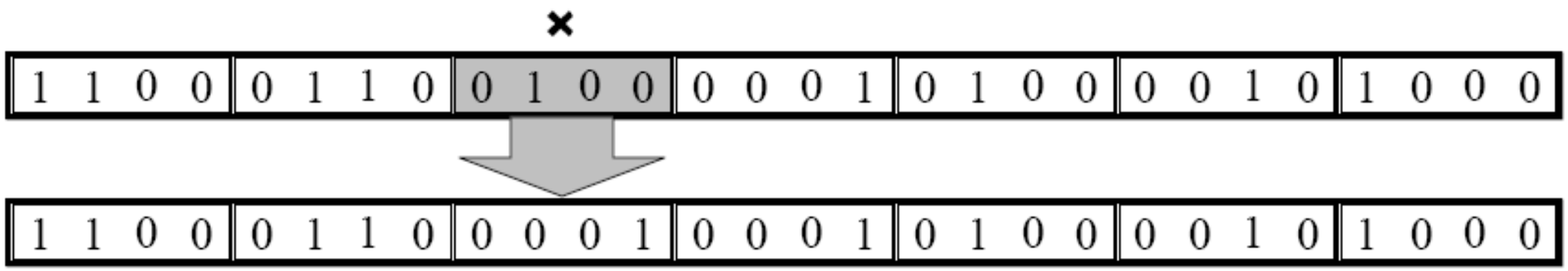
Maintenance Schedule: Mutation

- Mutation selects another gene from the allowed ‘gene pool’
  - No individual bit flipping



Unit 1:	1 1 0 0	0 1 1 0	0 0 1 1	
Unit 2:	1 1 0 0	0 1 1 0	0 0 1 1	
Unit 3:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 4:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 5:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 6:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1
Unit 7:	1 0 0 0	0 1 0 0	0 0 1 0	0 0 0 1

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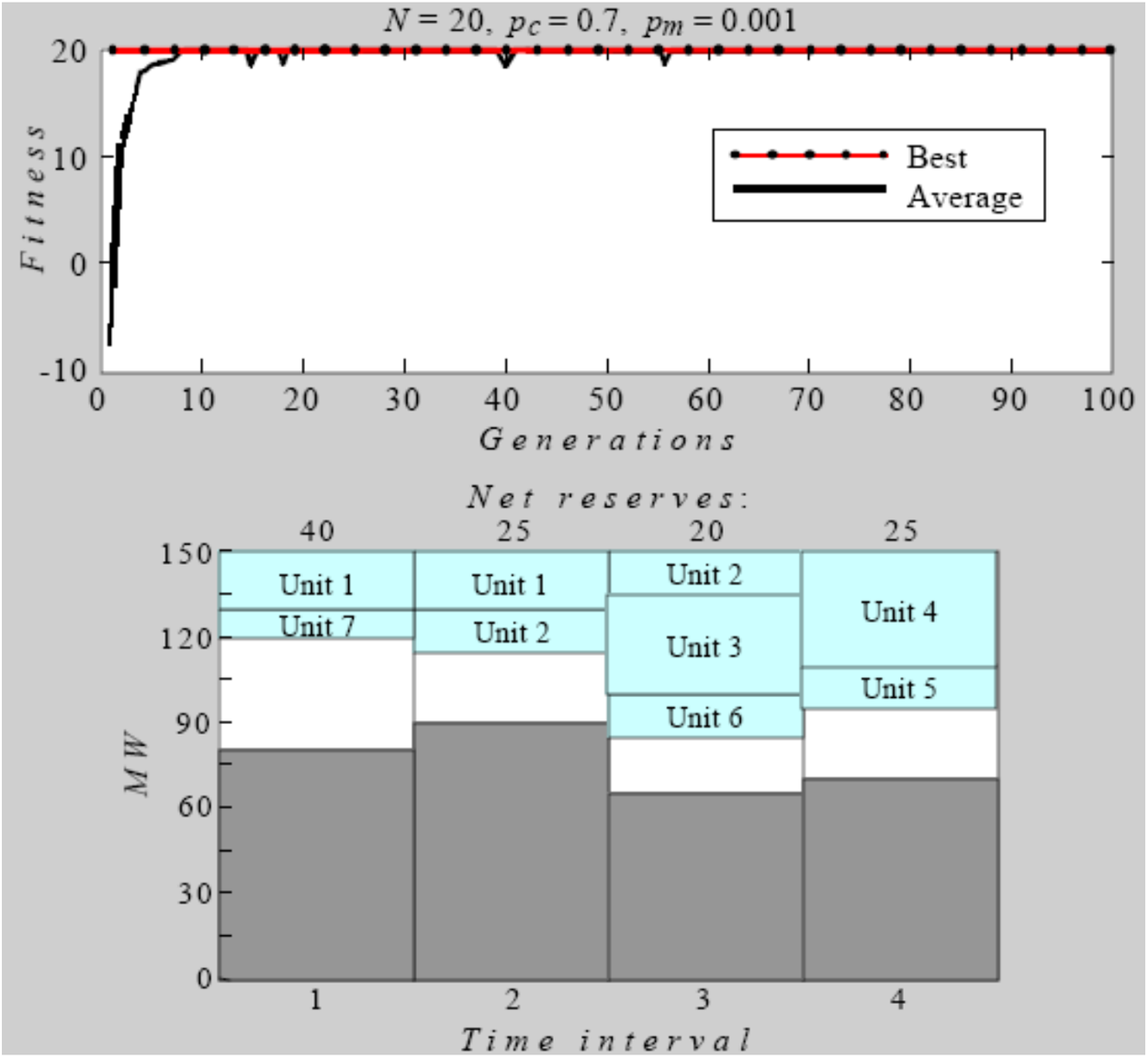
Maintenance Schedule



Population:
N = 20

Crossover prob:
$P_c = 0.7$

Mutation prob:
$P_m = 0.001$

Generations:
i = 100

Correction: Given the fitness function on slide 21, we cannot have negative average fitness as the performance graph above shows in the initial iterations.

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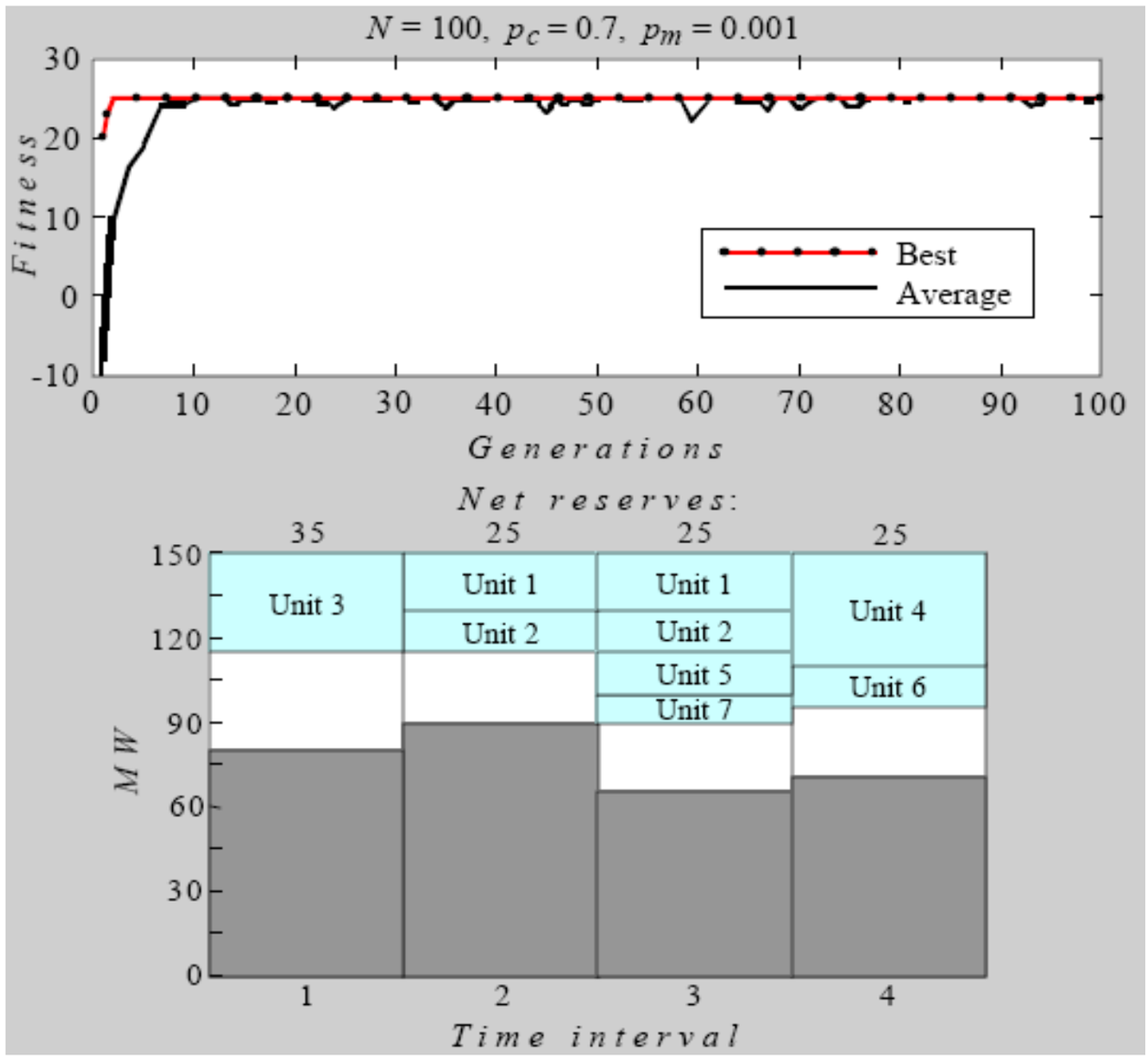
Maintenance Schedule: Performance



Population:
N = 100

Crossover prob:
$P_c = 0.7$

Mutation prob:
$P_m = 0.001$

Generations:
i = 100

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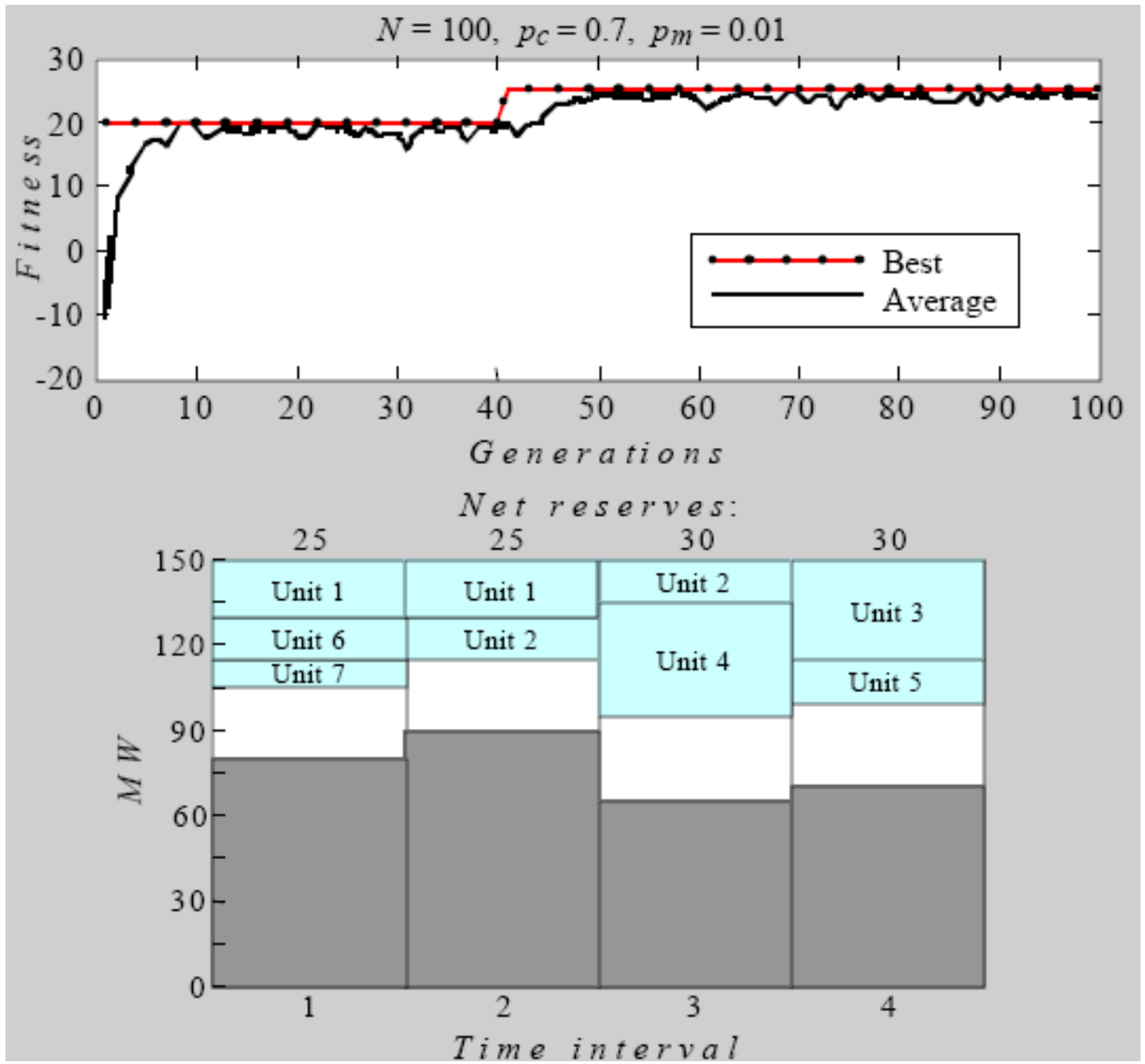
Maintenance Schedule: Performance



Population:
N = 100

Crossover prob:
$P_c = 0.7$

Mutation prob:
$P_m = 0.01$

Generations:
i = 100

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Multi-Objective Optimization with GA

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Why Multi-Objective Optimization?

Decisions often entail **multiple actors** with **conflicting interests**

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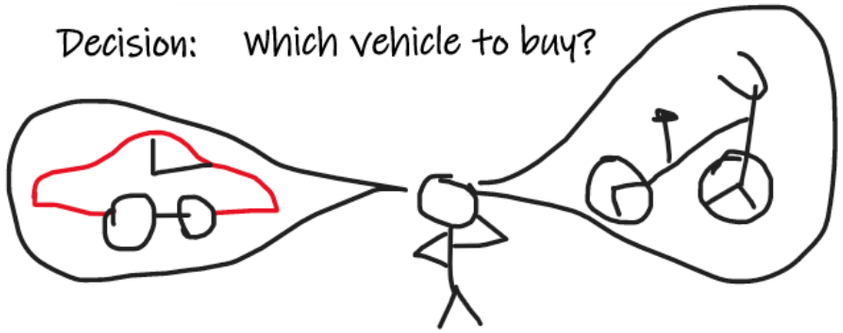
An Example Multi-Objective Problem

You are tasked to buy a vehicle.



To help guide your choices, answer the following questions:

1. What factors or goals do you consider when buying a vehicle?

2. How would you measure each of the factors or goals that you defined? In other words, how would you quantify how well does your decision meet those goals?

3. What constraints do you encounter when making that decision?

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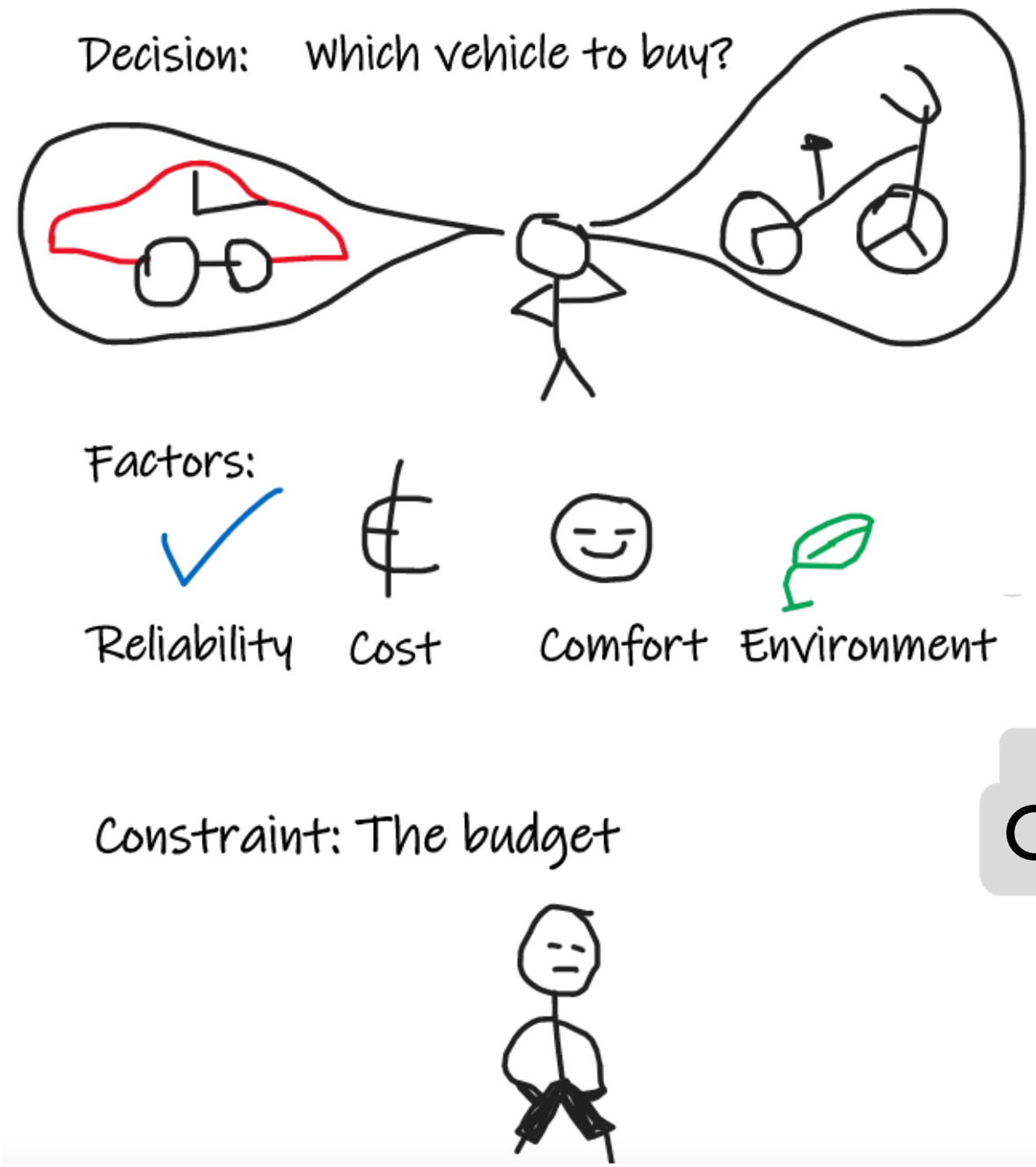
Defining a Multi-Objective Problem



A multi-objective problem consists of **two or more** objectives that conflict with each other

Alternative strategies that decision makers want to explore

Decisions

Objectives

Measures to rank the desirability of a certain decision

Constraints

Condition that needs to be satisfied often non-negotiable

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General Form of a Multi-Objective Optimization Problem

$$
\begin{aligned}
&\text{min/max} \quad f_m(\boldsymbol{x}), && m = 1, 2, \cdots, M \\
&\text{subject to } g_j(\boldsymbol{x}) \geq 0, && j = 1, 2, \cdots, J \\
&h_k(\boldsymbol{x}) = 0, && k = 1, 2, \cdots, K \\
&\underset{\text{lower bound}}{x_i^{(L)}} \leq x_i \leq \underset{\text{upper bound}}{x_i^{(U)}}, && i = 1, 2, \cdots, n
\end{aligned}
$$

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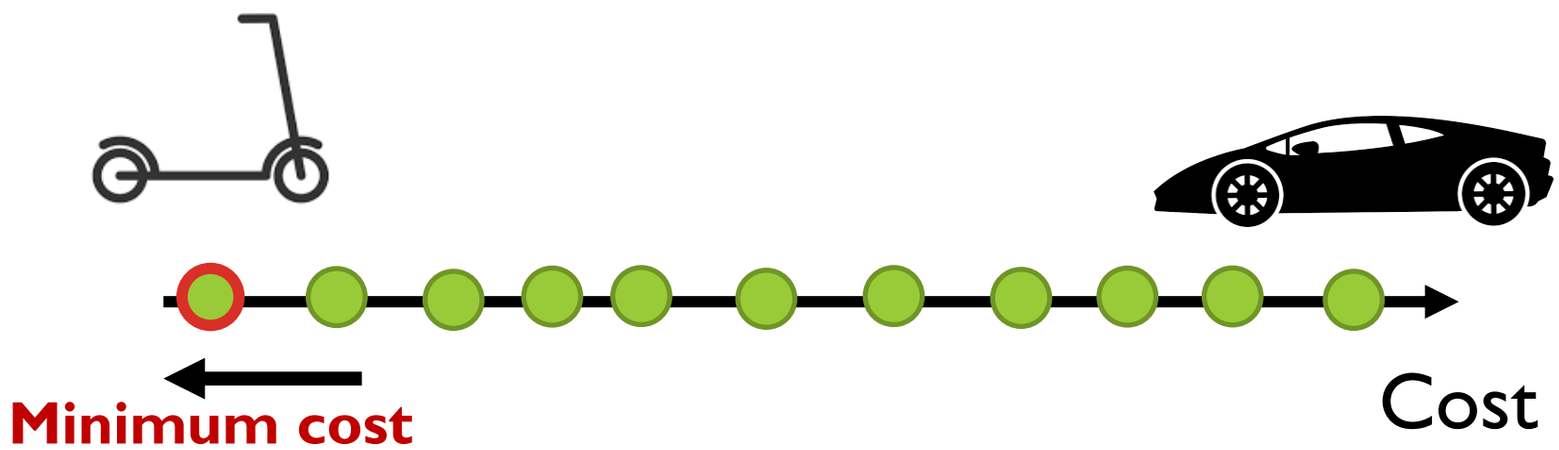
Solving Single Objective vs. Multi-Objective Problems

- **Single objective problem**:



- Find **a solution** that optimizes the given objective

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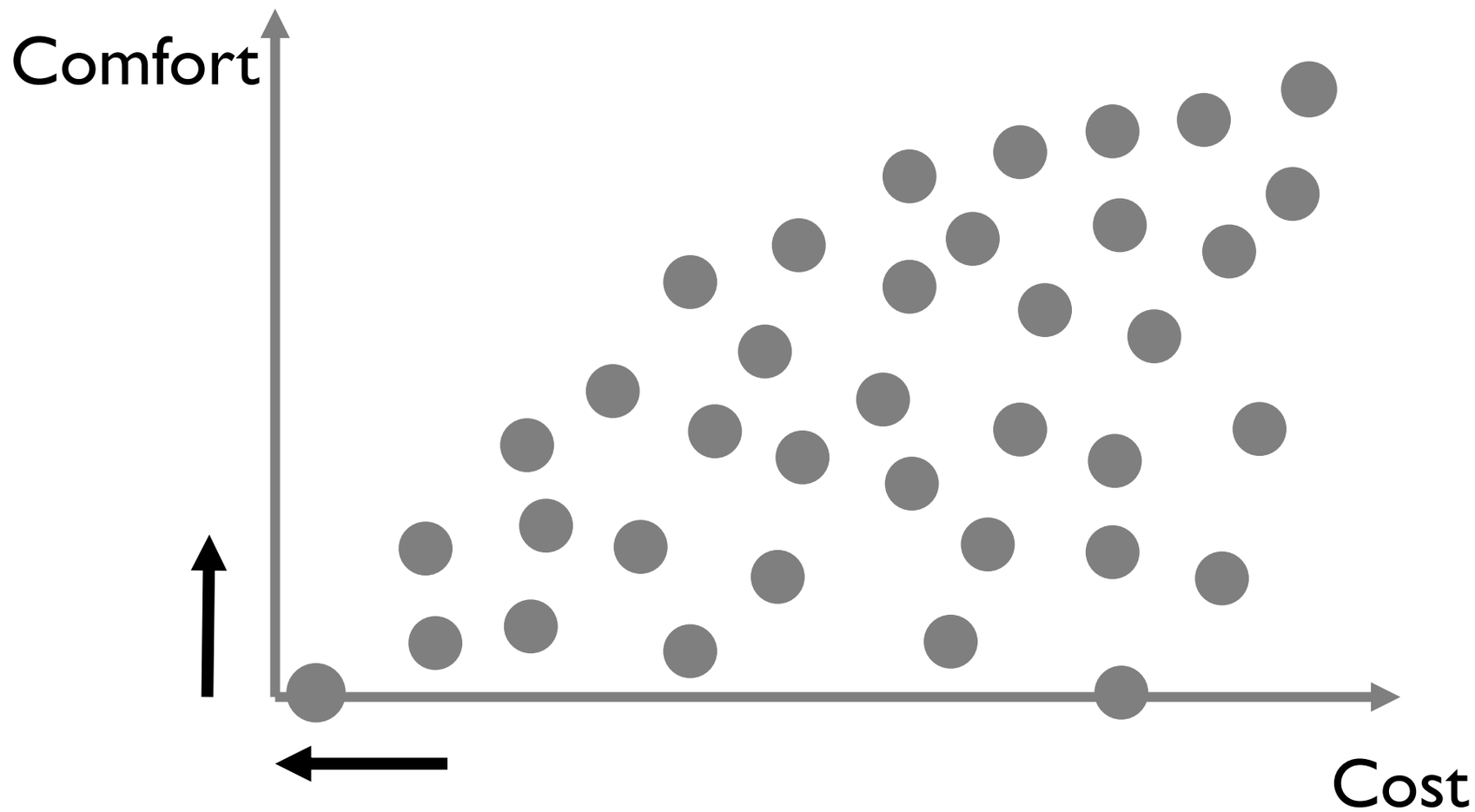
Solving Single Objective vs. Multi-Objective Problems

- **Multi-objective problem**:



- Find **the set of solutions** that define the best possible **tradeoffs** between competing objectives

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Pareto Optimality

- **Premise**:

  A solution is Pareto optimal when there exists no other solution that improves at least one objective without degrading the performance in any another objective

- **Purpose**:

  To rank alternatives in multi-objective problems

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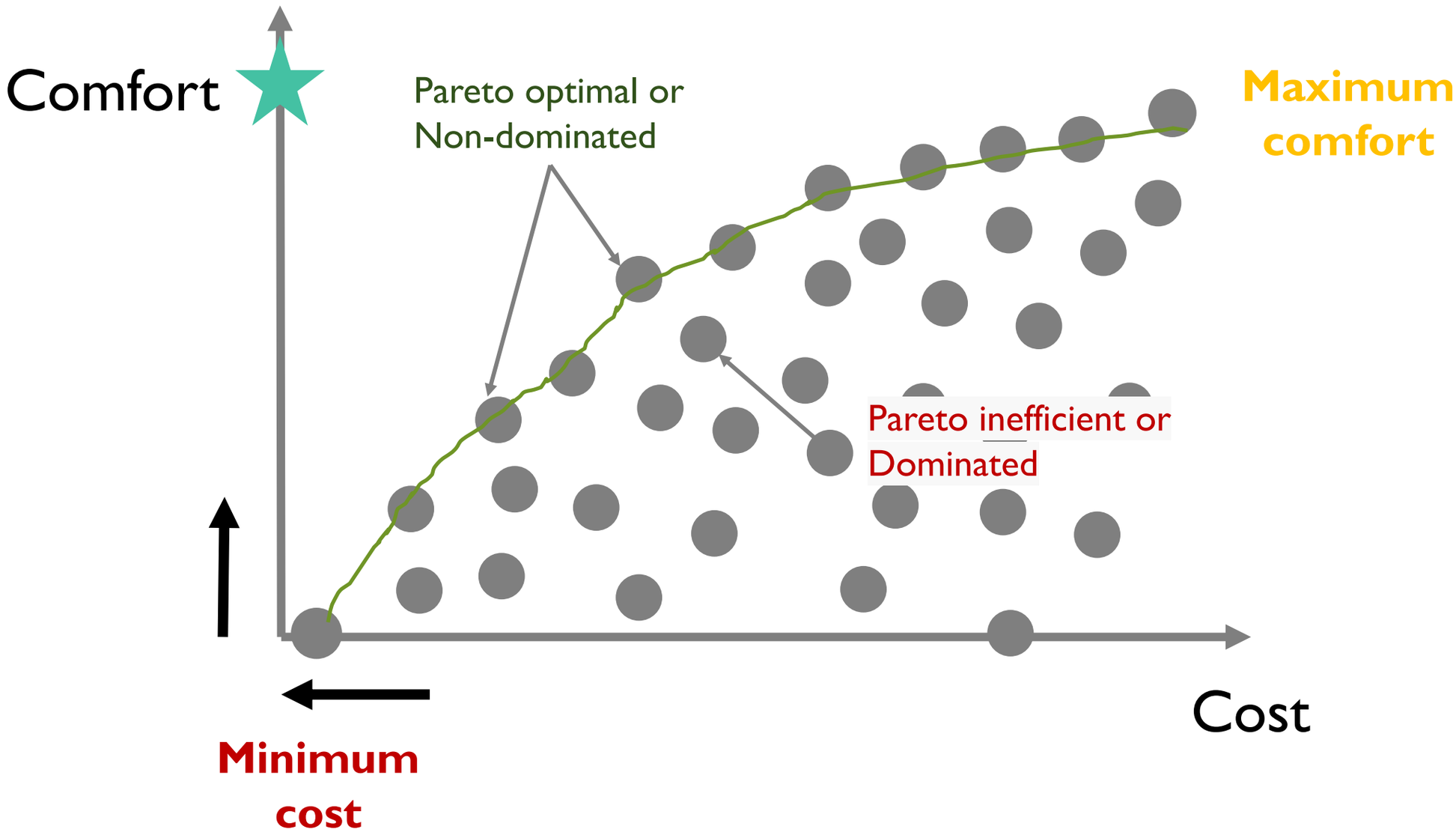
Pareto Optimal Front



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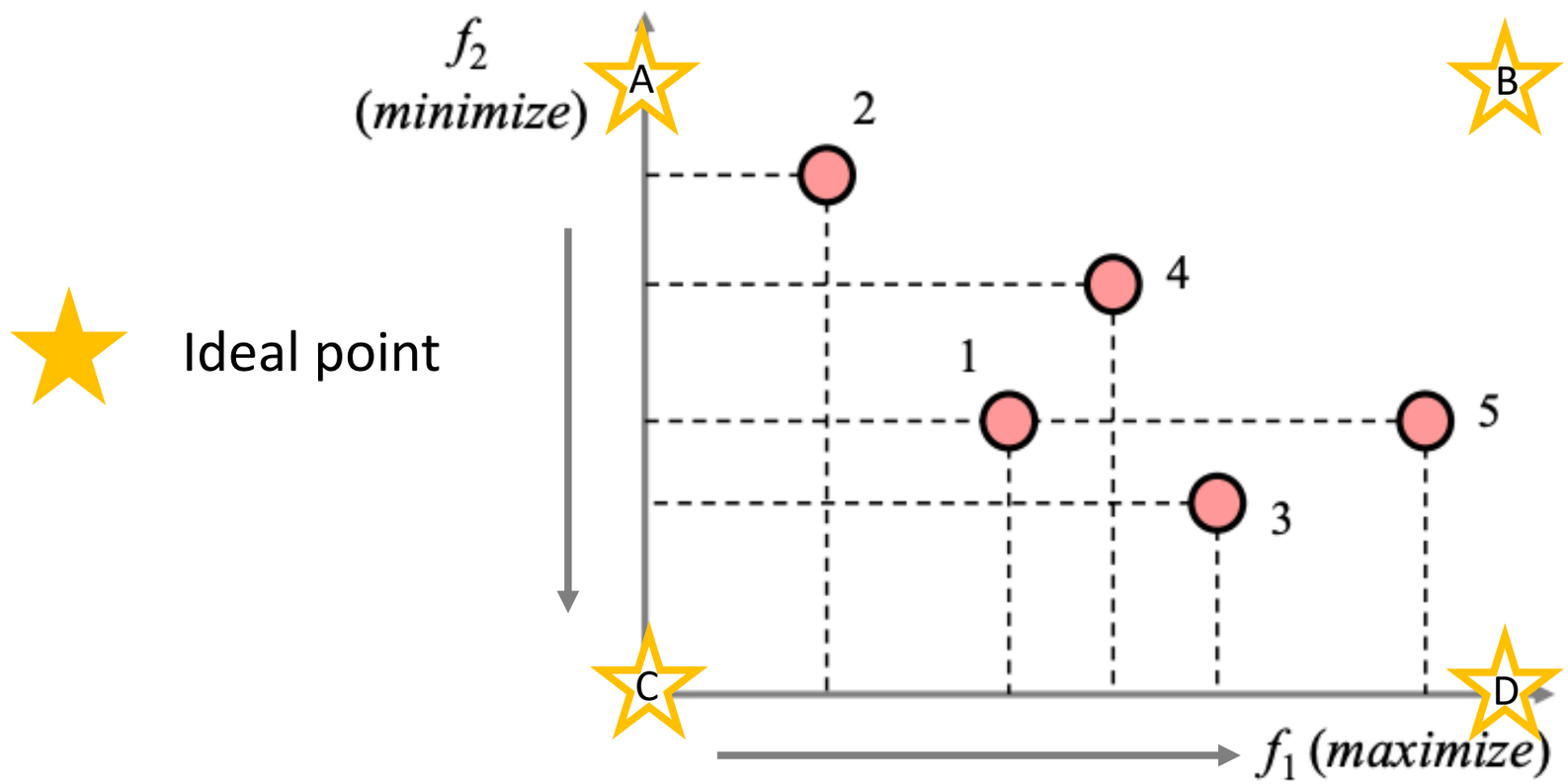
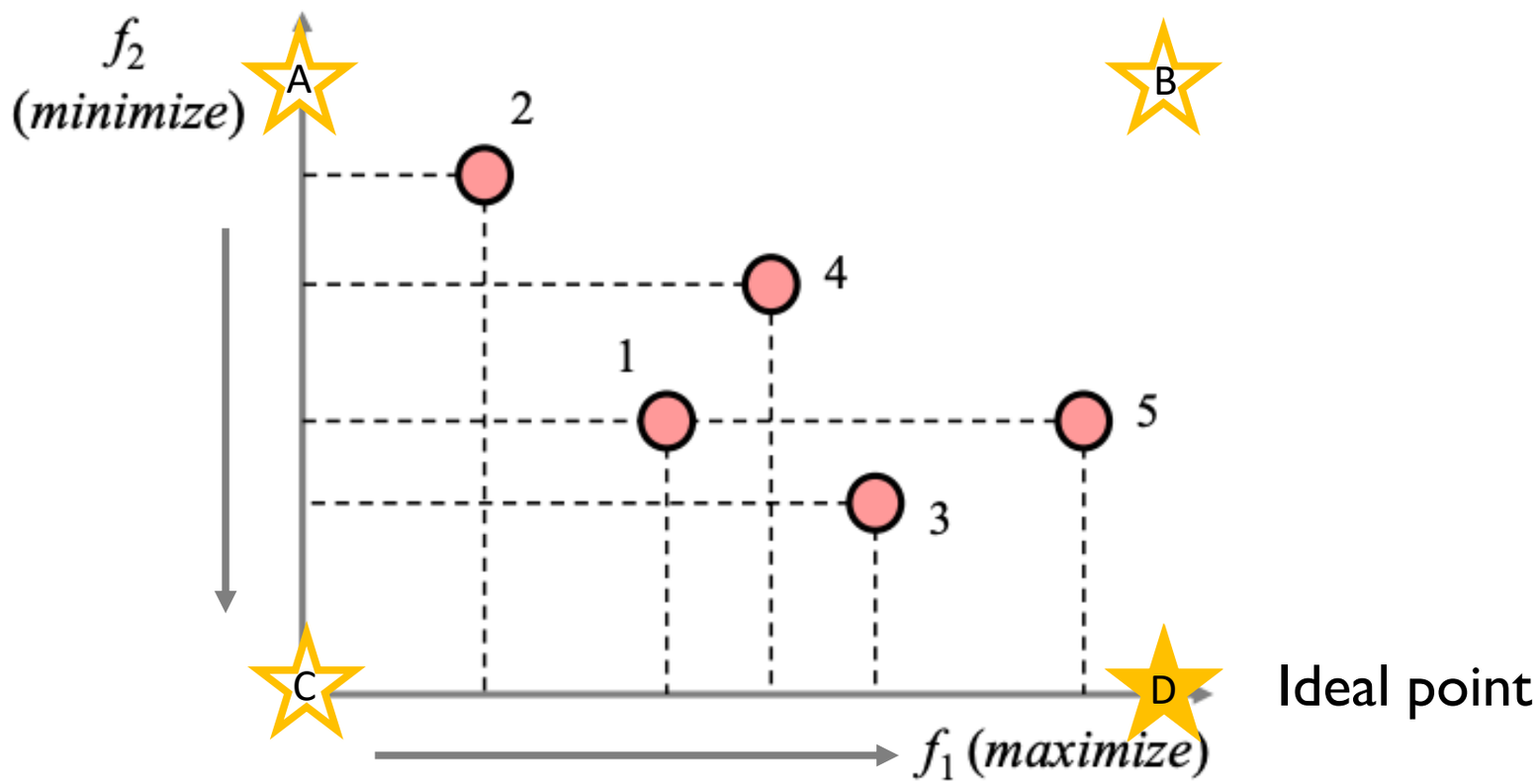
Pareto Dominance

Which of the following stars would be closer to the **ideal point**?



Pareto Dominance

Which of the following stars would be closer to the **ideal point**?



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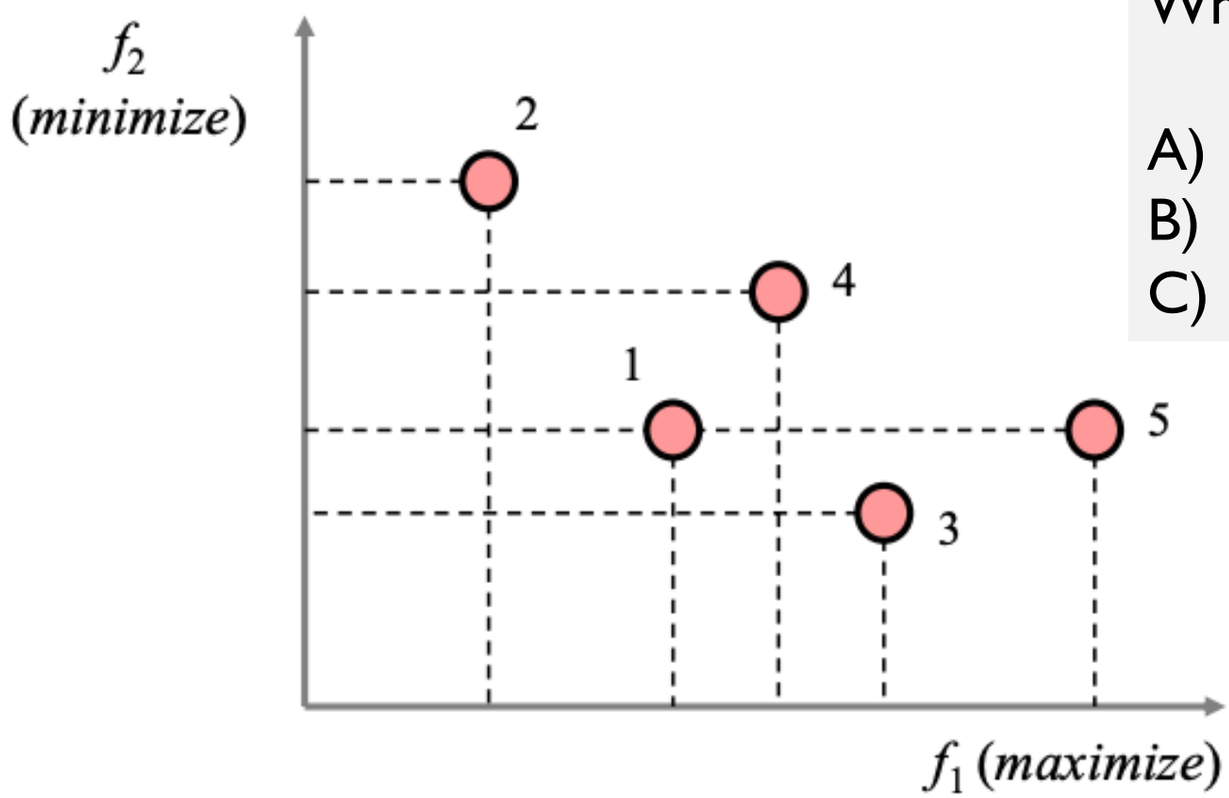
Dominance Test: 1 vs. 2



Which one dominates?

A) 1 dominates 2
B) 2 dominates 1
C) Neither dominates

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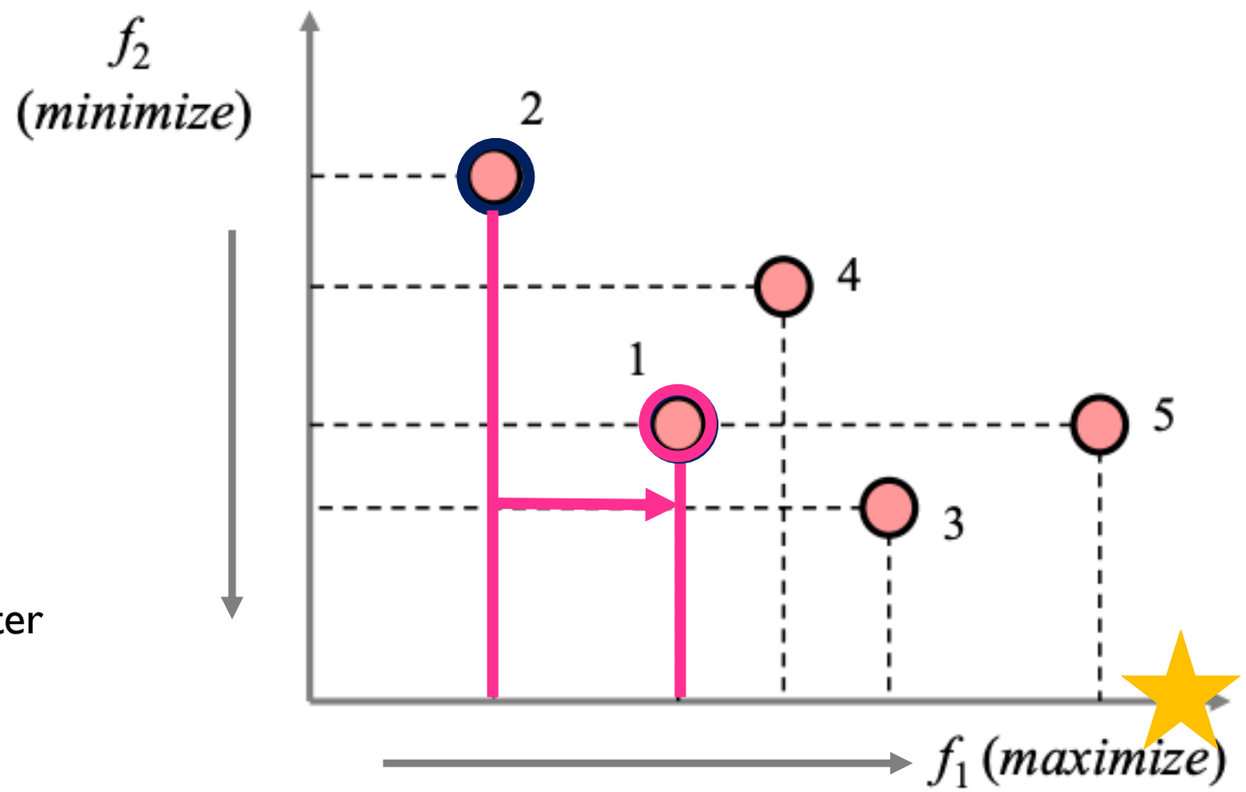
Dominance Test: 1 vs. 2



1 performs better than 2 for f_1

Ideal point

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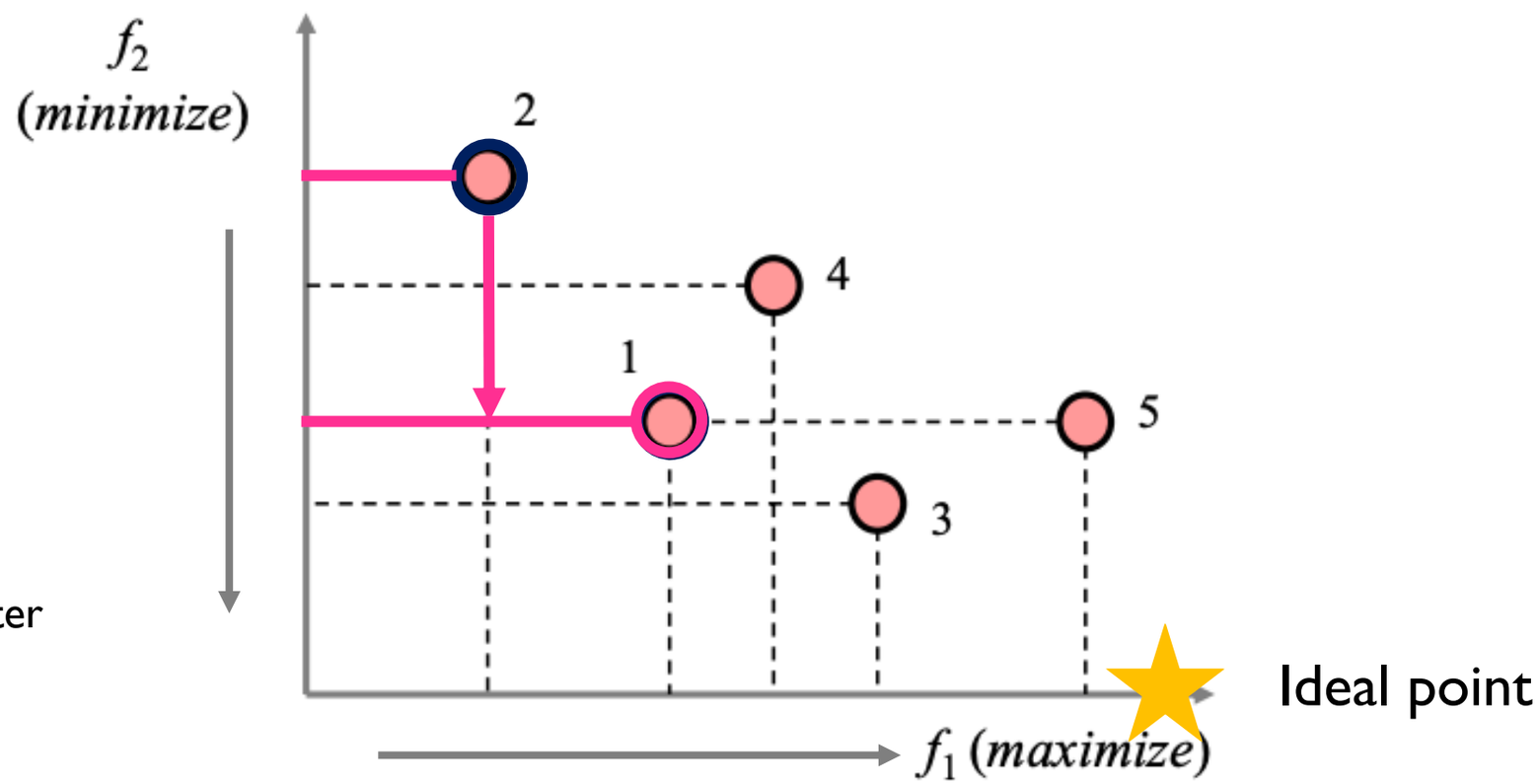
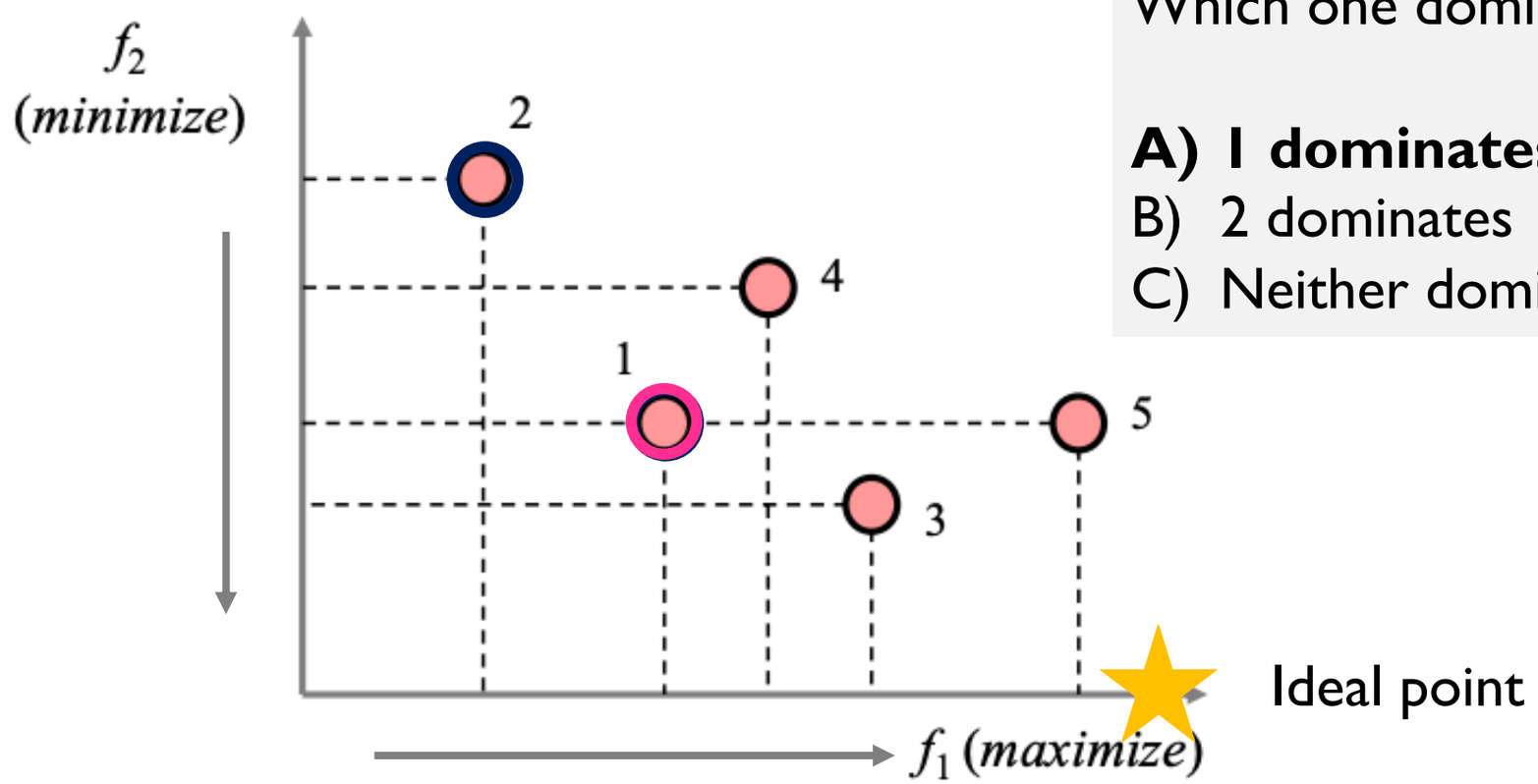
Dominance Test: 1 vs. 2



1 performs better than 2 for f_2

Dominance Test: 1 vs. 2



Which one dominates?

A) 1 dominates 2
B) 2 dominates 1
C) Neither dominates

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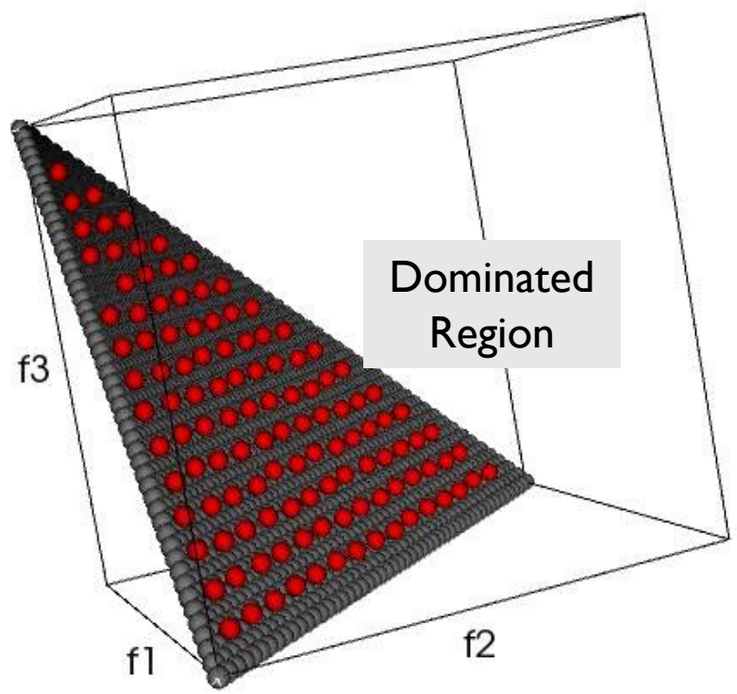
Pareto Optimal Surface

- Pareto optimality applies to more than two objectives as well



Three-objective test problem example

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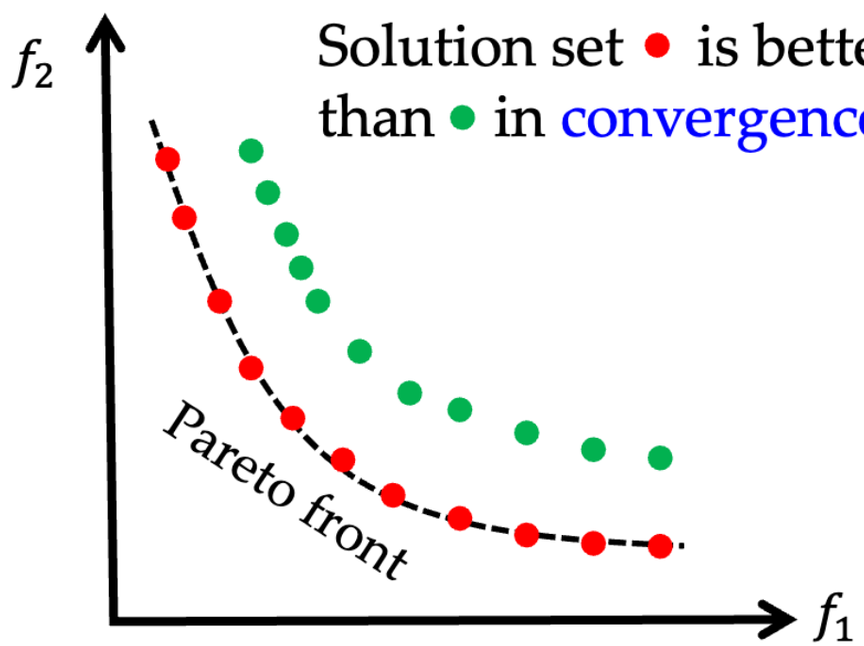
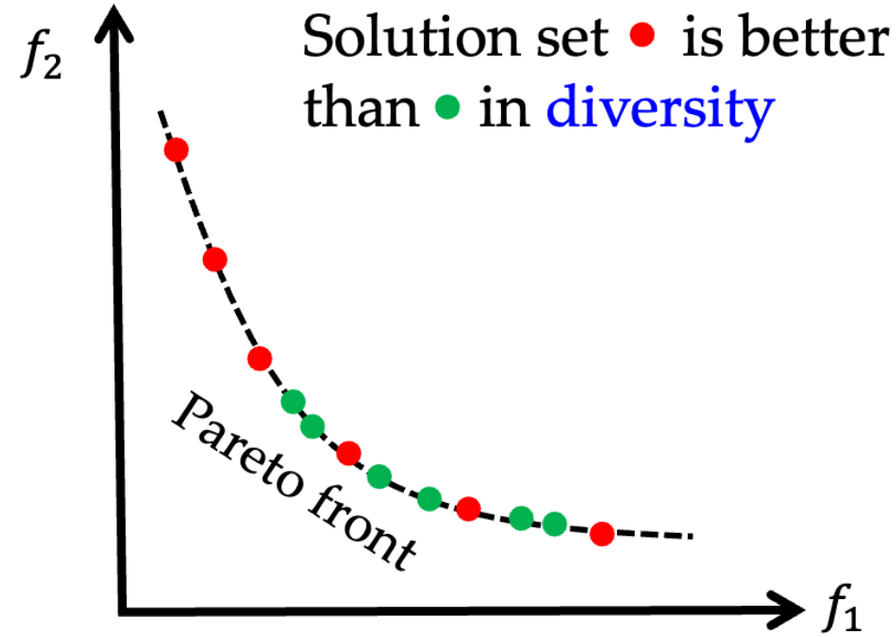
Quality of Pareto Optimality

Multi-objective optimization: optimize multiple objectives (which are usually conflicting) simultaneously

$$max_{x \in \mathcal{X}} \ \left(f_1(x), f_2(x), \ldots, f_m(x)\right)$$





Bi-objective minimization

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Goal of Optimization

- We are searching for Pareto optimal solutions. In general, the Pareto front consists of infinitely many points. The algorithm can only find a finite number of solutions.
- Anyway, a human decision maker can analyze only a rather small number of solutions.

- Hence a good MOO algorithm must output a set of limited size. It is not uncommon to specify a maximal size, e.g., 100 points.

- Many very similar solutions concentrated on a small area of the Pareto front are of limited value, since only few essentially different compromises are represented.

- Therefore we demand in addition that the finite solution set shall be as representative as possible, meaning that it should be evenly distributed over the Pareto front.

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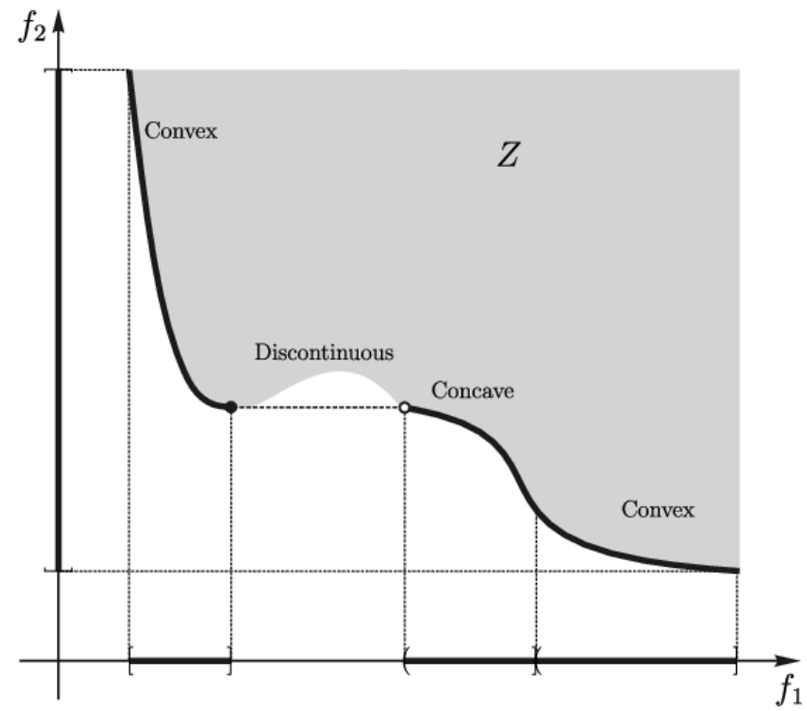
Why Use Genetic Algorithm for Multi-Objective Optimization



- **Non-linear** objective functions
- Insensitive to **discontinuities** or **shape** of the Pareto front.
- **Flexible**, can find solutions to problems for which we have little knowledge
- The Pareto set is generated in a **single run**

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Key difference

- Evolutionary algorithms for multi-objective optimization often use their population for the representation of the solution set.
- This gives a new role to the population. When optimizing a single objective, the population acts as a pool of building blocks from which the variation operators should be able to create improvements.
- In the multi-objective case, each individual must perform well, not only the best one, since the whole front must be covered, not only a single optimum.
- This is taken into account by MOO specific selection mechanisms.

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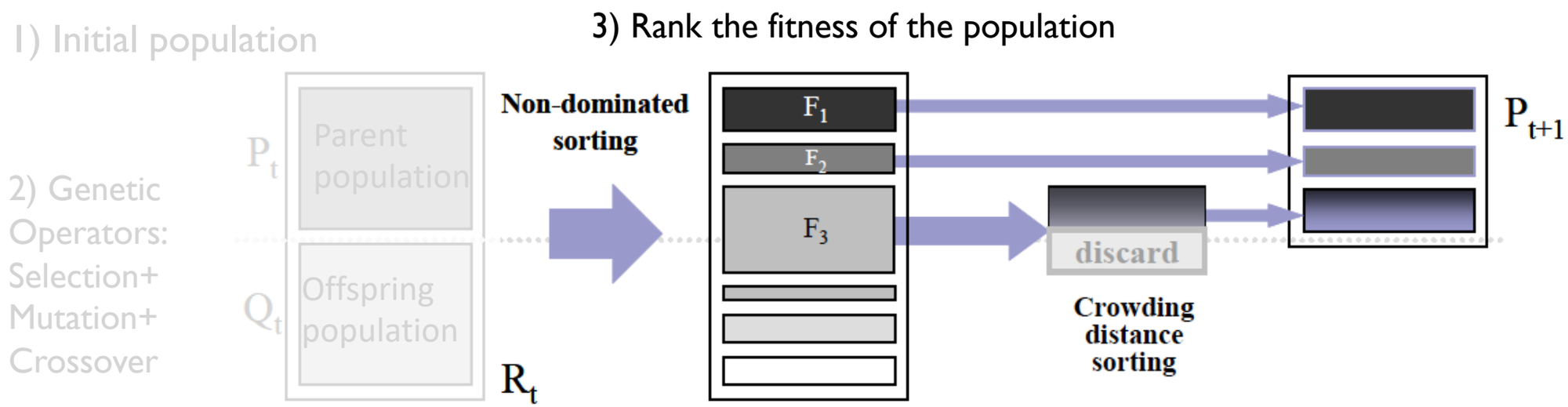
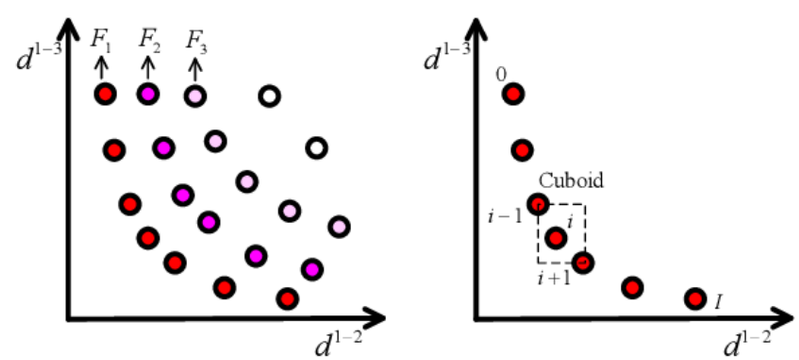
NSGA II

NSGA II: Non-dominated Sorting Genetic Algorithm (Deb et al. 2000)





(a) Non-dominated sorting (b) Crowding-distance computation

K. Deb, A. Pratap, S. Agarwal and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," in *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182-197, April 2002, doi: 10.1109/4235.996017.

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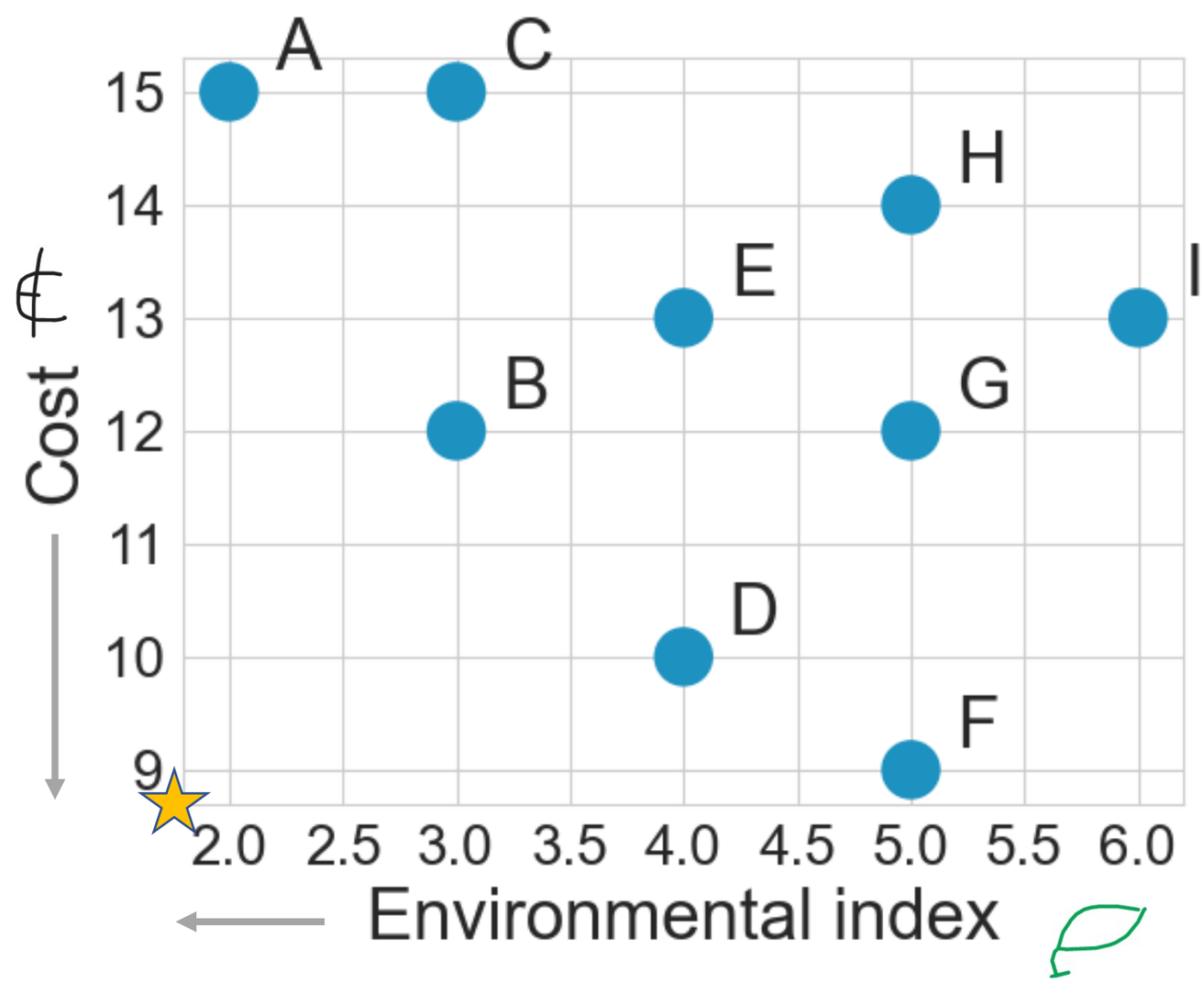
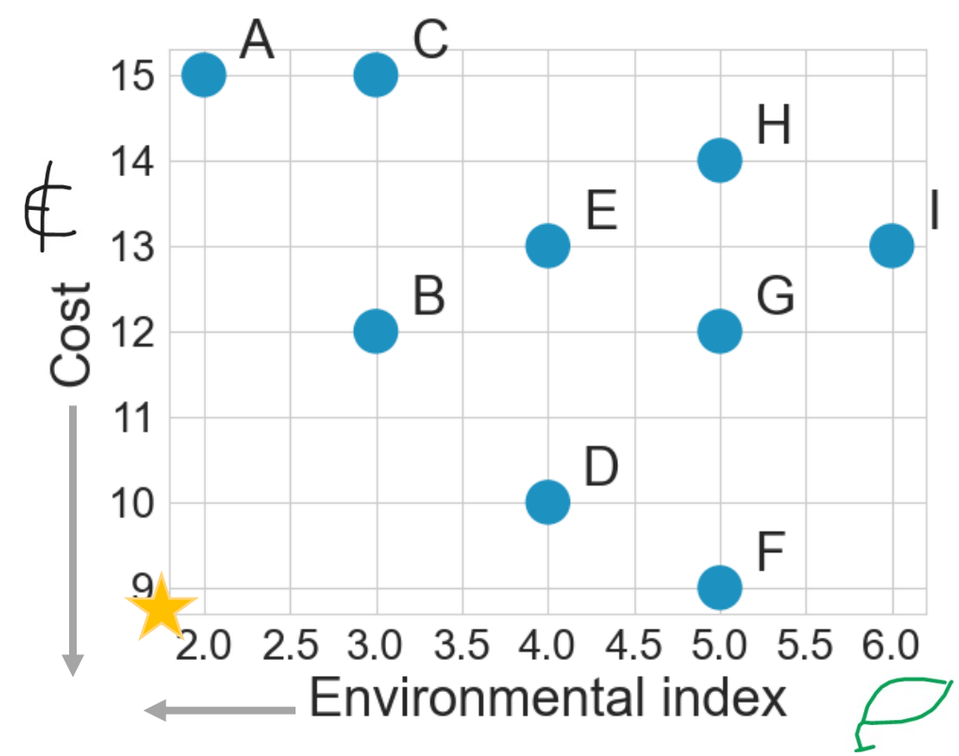
NSGA II: Example

Car type	Environmental index (minimize)	Cost (minimize)
A	2	15K
B	3	12K
C	3	15K
D	4	10K
E	4	13K
F	5	9K
G	5	12K
H	5	14K
I	6	13K



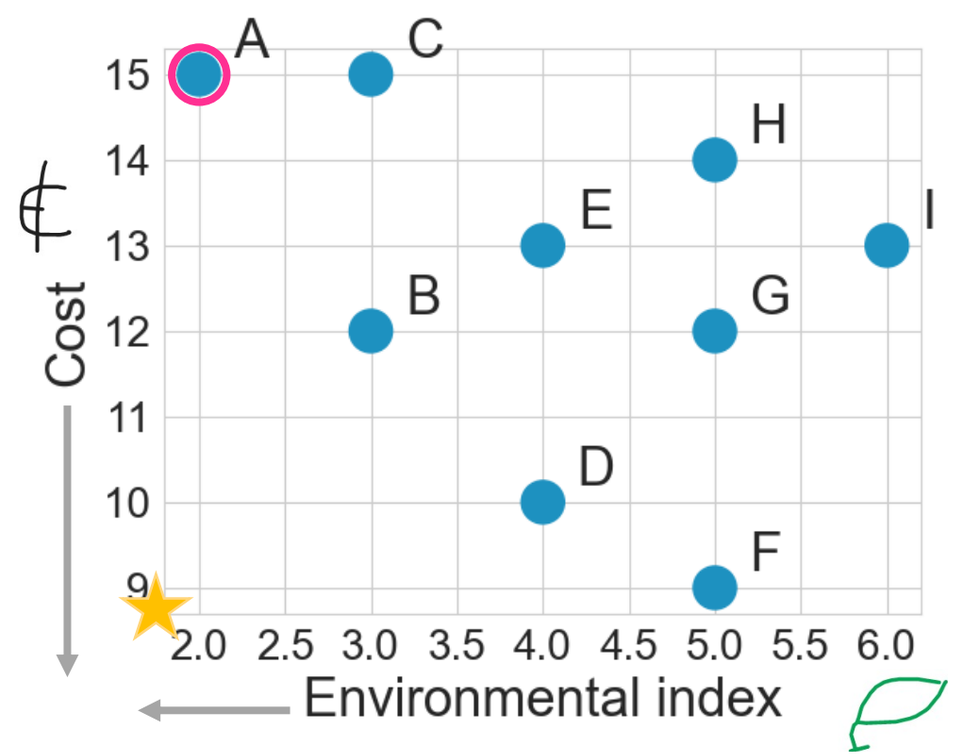
NSGA II Example: Non-dominated Sorting

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate
A	2	15K		
B	3	12K		
C	3	15K		
D	4	10K		
E	4	13K		
F	5	9K		
G	5	12K		
H	5	14K		
I	6	13K		



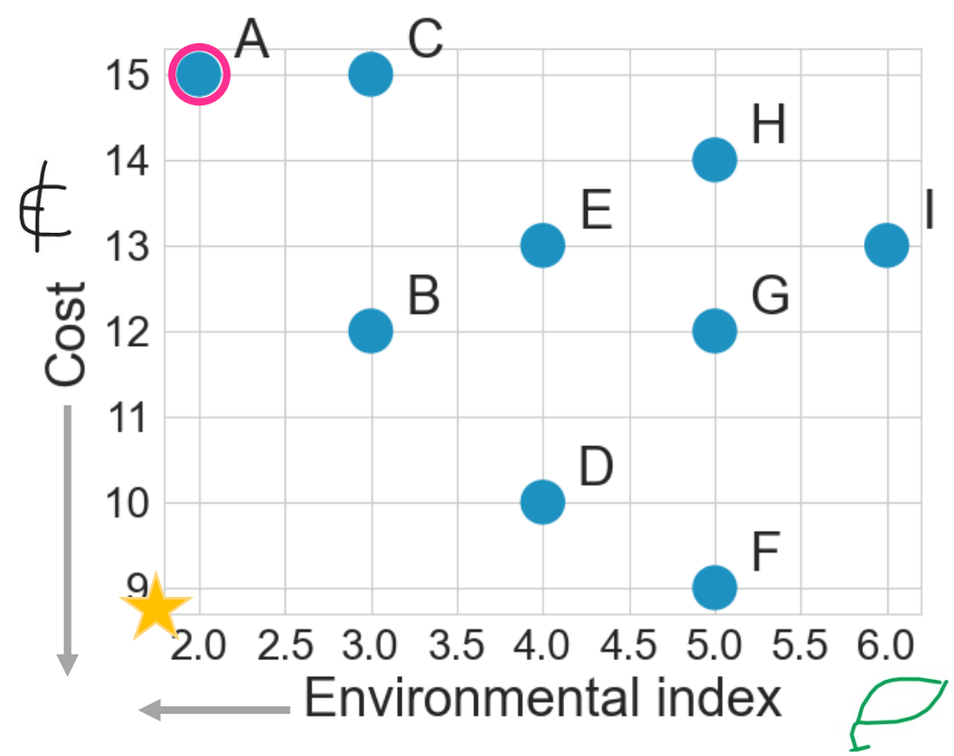
NSGA II Example: Non-dominated Sorting

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate
A	**2**	**15K**		
B	3	12K		
C	3	15K		
D	4	10K		
E	4	13K		
F	5	9K		
G	5	12K		
H	5	14K		
I	6	13K		



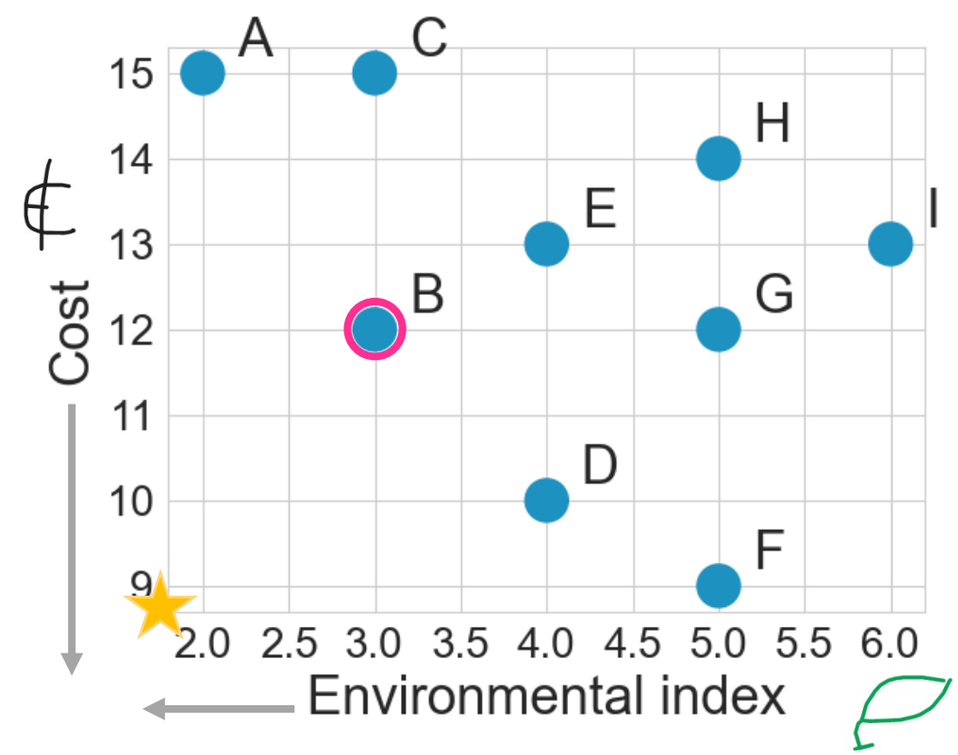
NSGA II Example: Non-dominated Sorting

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate
A	**2**	**15K**	**C**	**0**
B	3	12K		
C	3	15K		
D	4	10K		
E	4	13K		
F	5	9K		
G	5	12K		
H	5	14K		
I	6	13K		



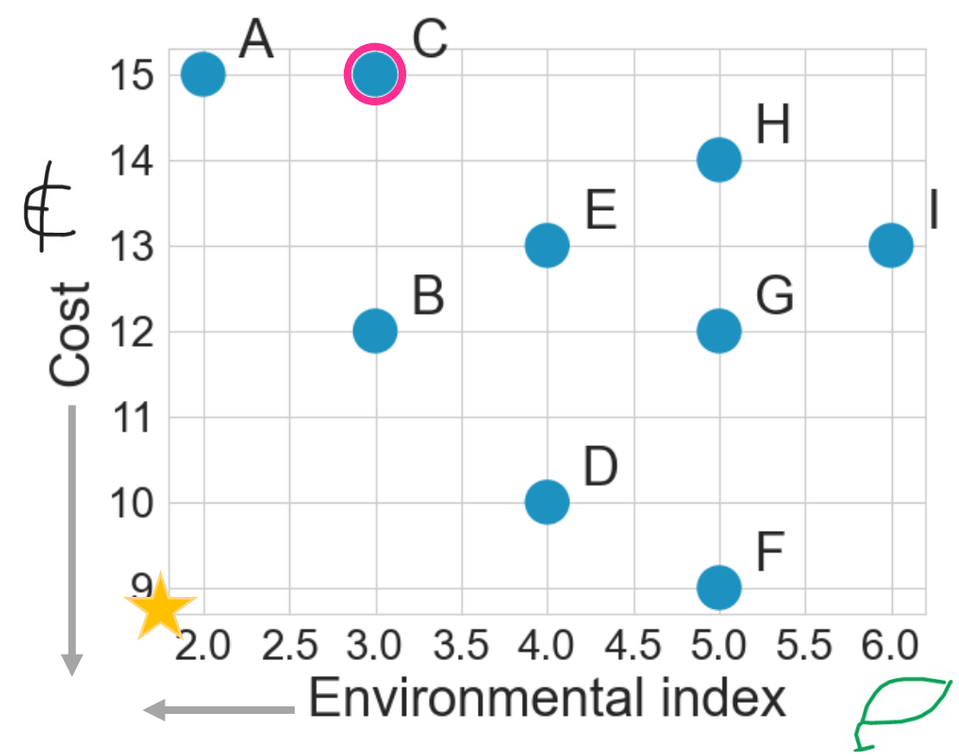
NSGA II Example: Non-dominated Sorting

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate
A	2	15K	C	0
B	**3**	**12K**	**C,E,G,H,I**	**0**
C	3	15K		
D	4	10K		
E	4	13K		
F	5	9K		
G	5	12K		
H	5	14K		
I	6	13K		



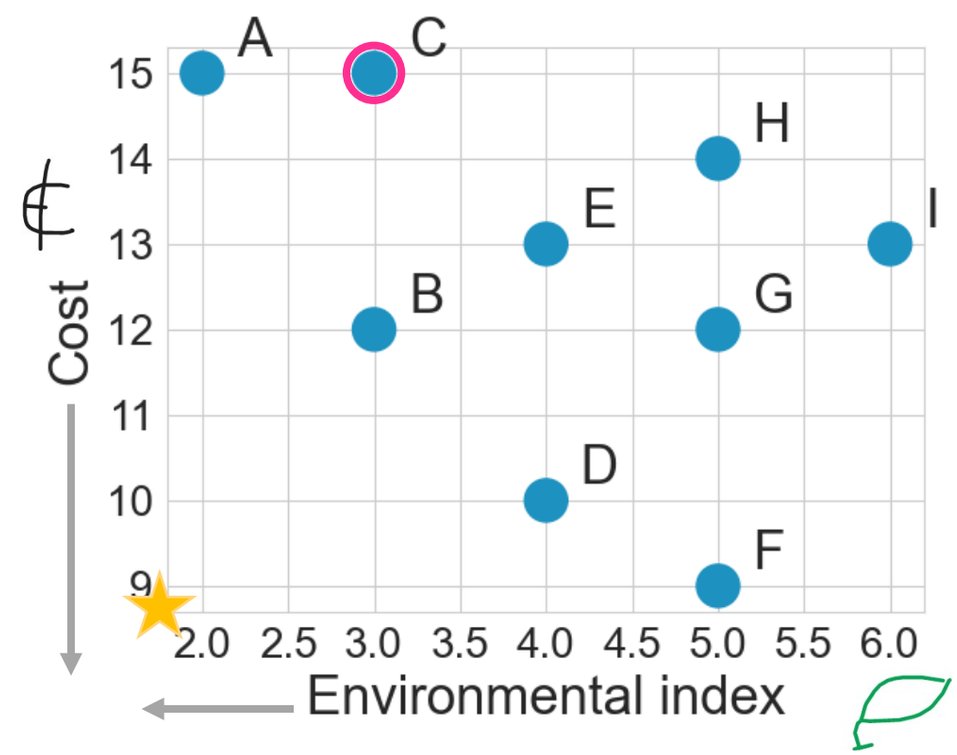
NSGA II Example: Non-dominated Sorting

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate
A	2	15K	C	0
B	3	12K	C,E,G,H,I	0
C	**3**	**15K**		
D	4	10K		
E	4	13K		
F	5	9K		
G	5	12K		
H	5	14K		
I	6	13K		



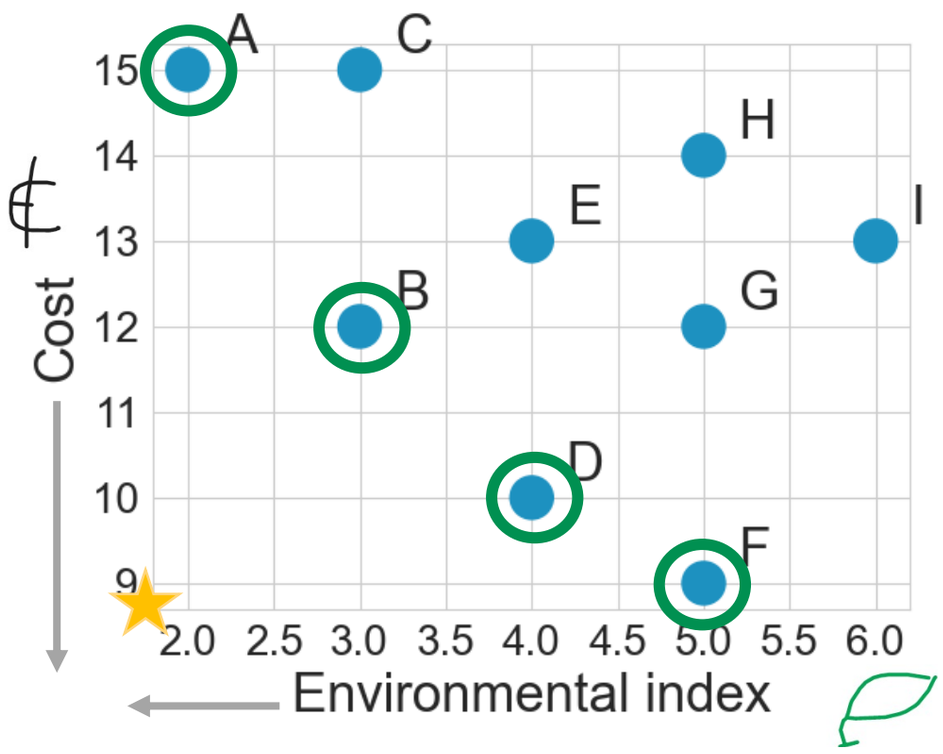
NSGA II Example: Non-dominated Sorting

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate
A	2	15K	C	0
B	3	12K	C,E,G,H,I	0
C	**3**	**15K**	**-**	**2**
D	4	10K		
E	4	13K		
F	5	9K		
G	5	12K		
H	5	14K		
I	6	13K		



NSGA II Example: Pareto Fronts

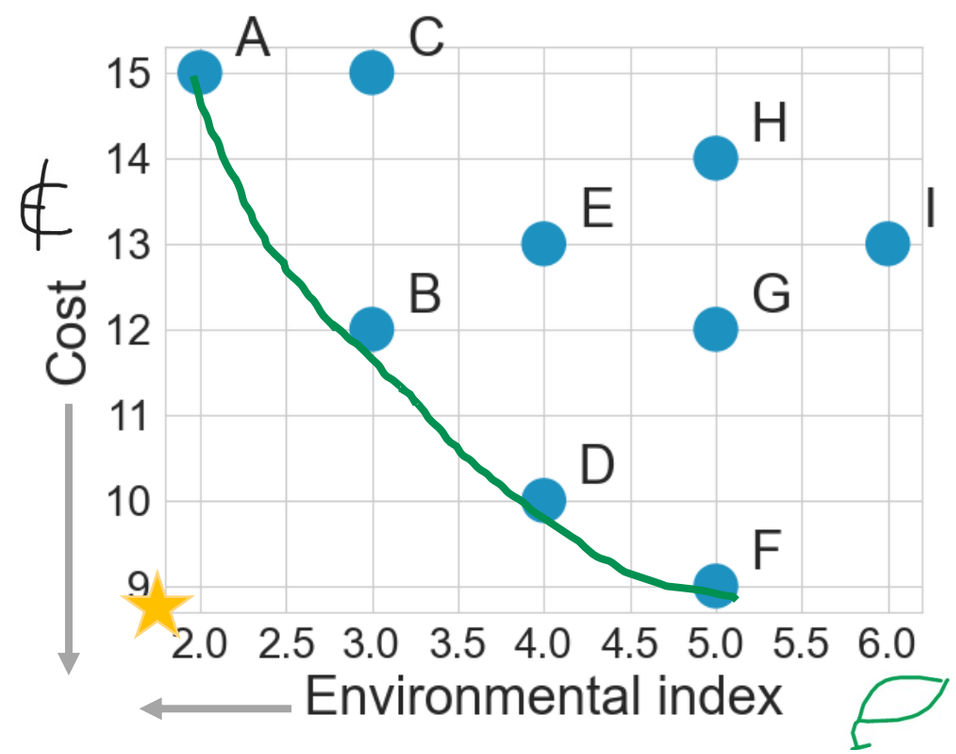
Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	2	15K	C	0	
B	3	12K	C,E,G,H,I	0	
C	3	15K	-	2	
D	4	10K	E,G,H,I	0	
E	4	13K	H,I	2	
F	5	9K	G,H,I	0	
G	5	12K	H,I	3	
H	5	14K	-	5	
I	6	13K	-	5	



NSGA II Example: Pareto Fronts

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	**2**	**15K**	**C**	**0**	**1**
B	**3**	**12K**	**C,E,G,H,I**	**0**	**1**
C	3	15K	-	2	
D	**4**	**10K**	**E,G,H,I**	**0**	**1**
E	4	13K	H,I	2	
F	**5**	**9K**	**G,H,I**	**0**	**1**
G	5	12K	H,I	3	
H	5	14K	-	5	
I	6	13K	-	5	

Non-dominated front 1

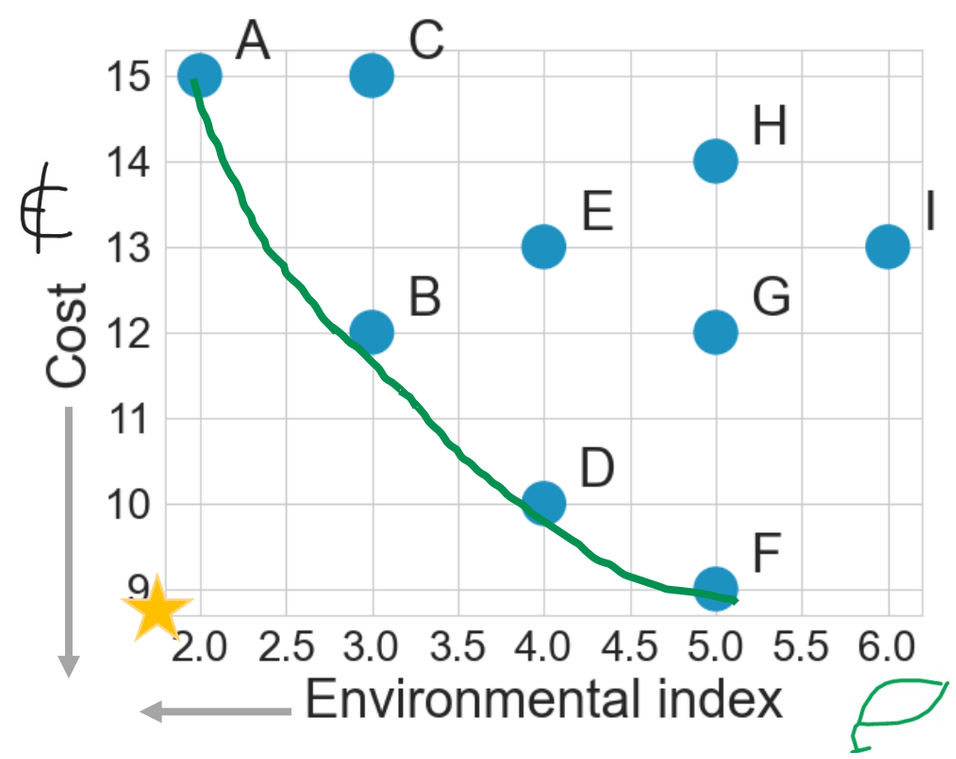


NSGA II Example: Pareto Fronts

Update "No. points that dominate" by taking out the identified non-dominated front

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	**2**	**15K**	**C**	**0**	**1**
B	**3**	**12K**	**C,E,G,H,I**	**0**	**1**
C	3	15K	-	2-1-1	
D	**4**	**10K**	**E,G,H,I**	**0**	**1**
E	4	13K	H,I	2-1-1	
F	**5**	**9K**	**G,H,I**	**0**	**1**
G	5	12K	H,I	3-1-1-1	
H	5	14K	-	5-1-1-1	
I	6	13K	-	5-1-1-1	

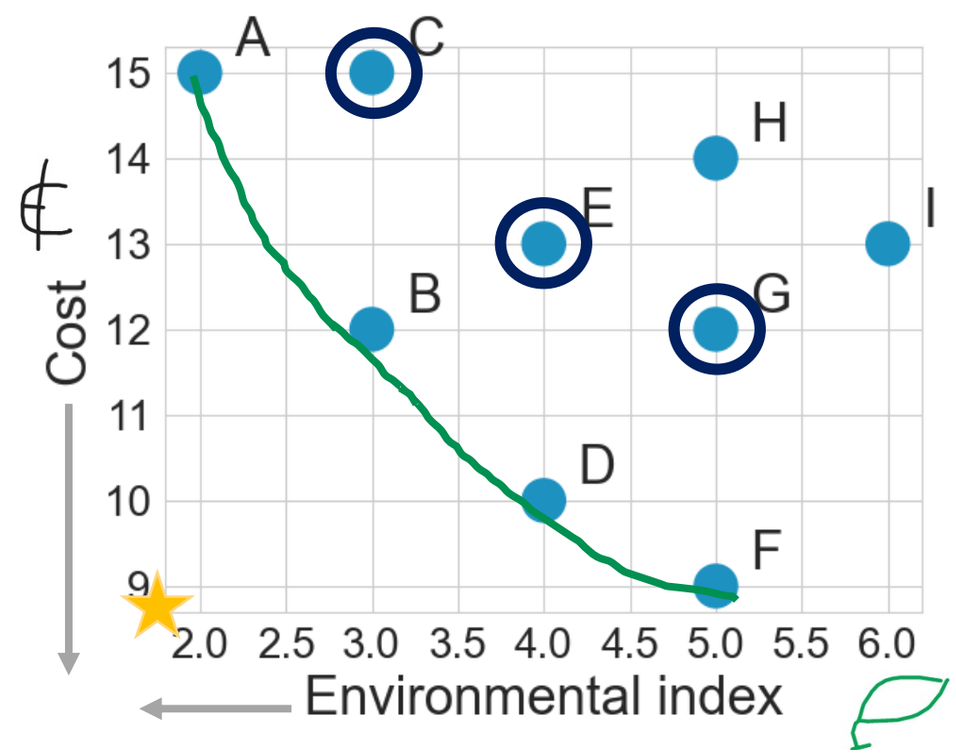
Non-dominated front 1



NSGA II Example: Pareto Fronts

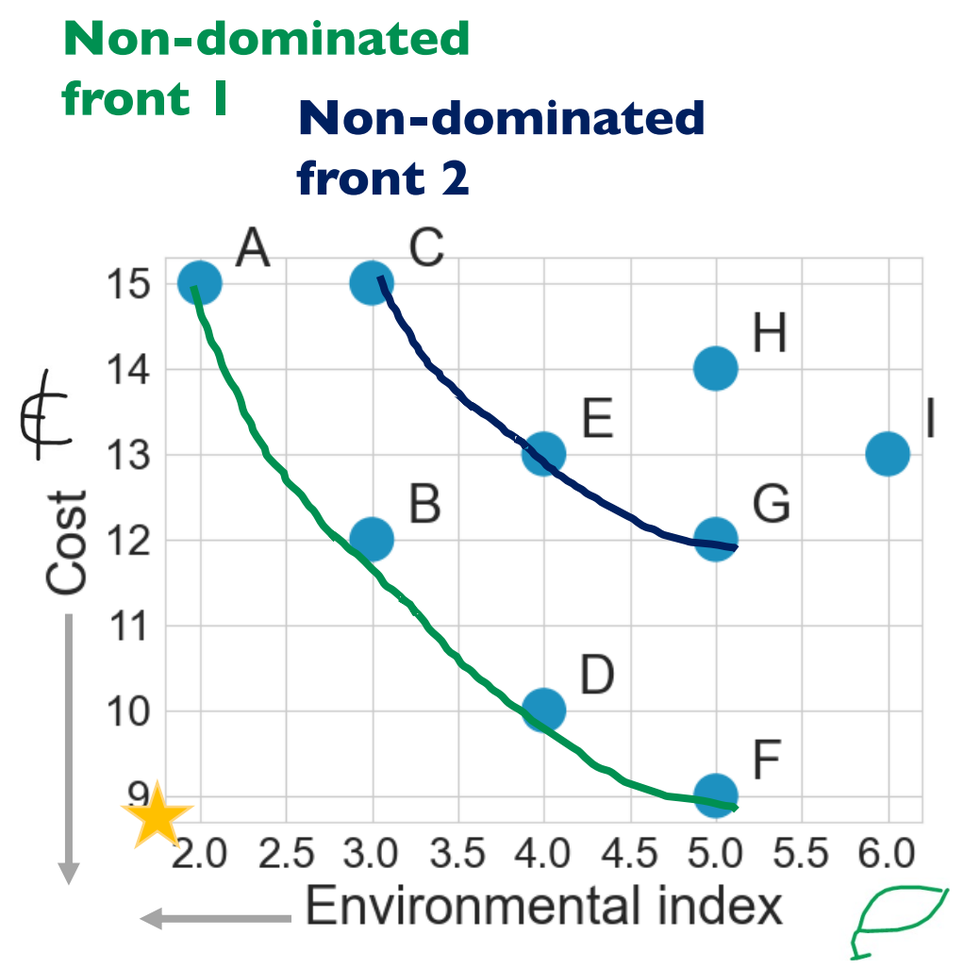
Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	**2**	**15K**	**C**	**0**	**1**
B	**3**	**12K**	**C,E,G,H,I**	**0**	**1**
C	3	15K	-	0	
D	**4**	**10K**	**E,G,H,I**	**0**	**1**
E	4	13K	H,I	0	
F	**5**	**9K**	**G,H,I**	**0**	**1**
G	5	12K	H,I	0	
H	5	14K	-	2	
I	6	13K	-	2	

Non-dominated front 1



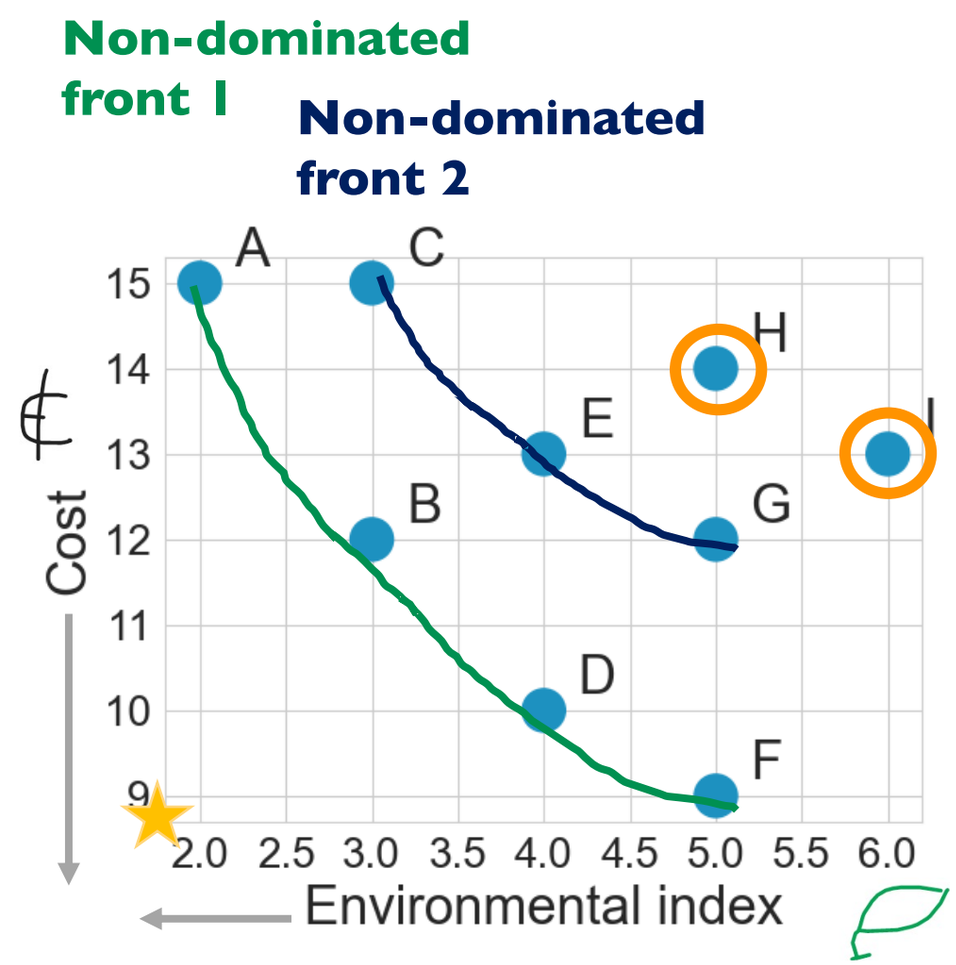
NSGA II Example: Pareto Fronts

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	2	15K	C	0	1
B	3	12K	C,E,G,H,I	0	1
C	3	15K	-	0	2
D	4	10K	E,G,H,I	0	1
E	4	13K	H,I	0	2
F	5	9K	G,H,I	0	1
G	5	12K	H,I	0	2
H	5	14K	-	2-1-1	
I	6	13K	-	2-1-1	



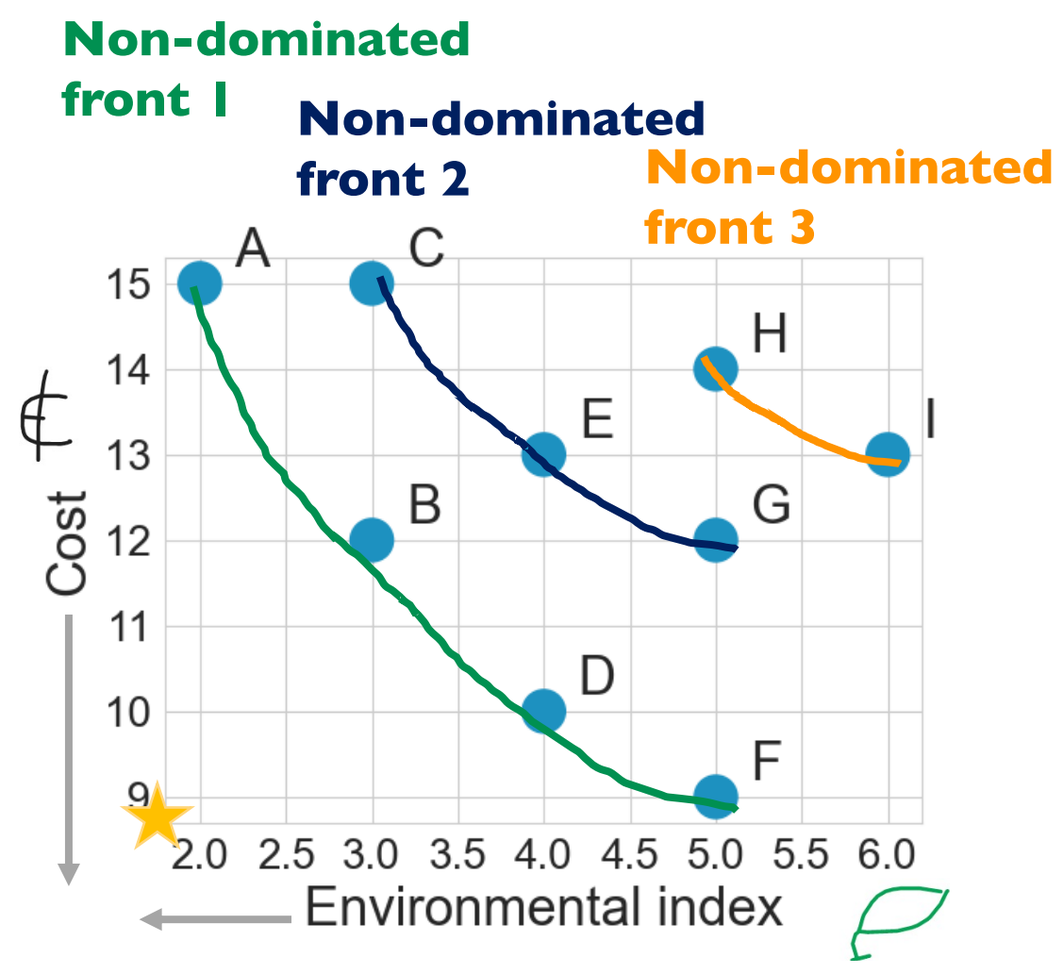
NSGA II Example: Pareto Fronts

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	2	15K	C	0	1
B	3	12K	C,E,G,H,I	0	1
C	3	15K	-	0	2
D	4	10K	E,G,H,I	0	1
E	4	13K	H,I	0	2
F	5	9K	G,H,I	0	1
G	5	12K	H,I	0	2
H	5	14K	-	0	
I	6	13K	-	0	

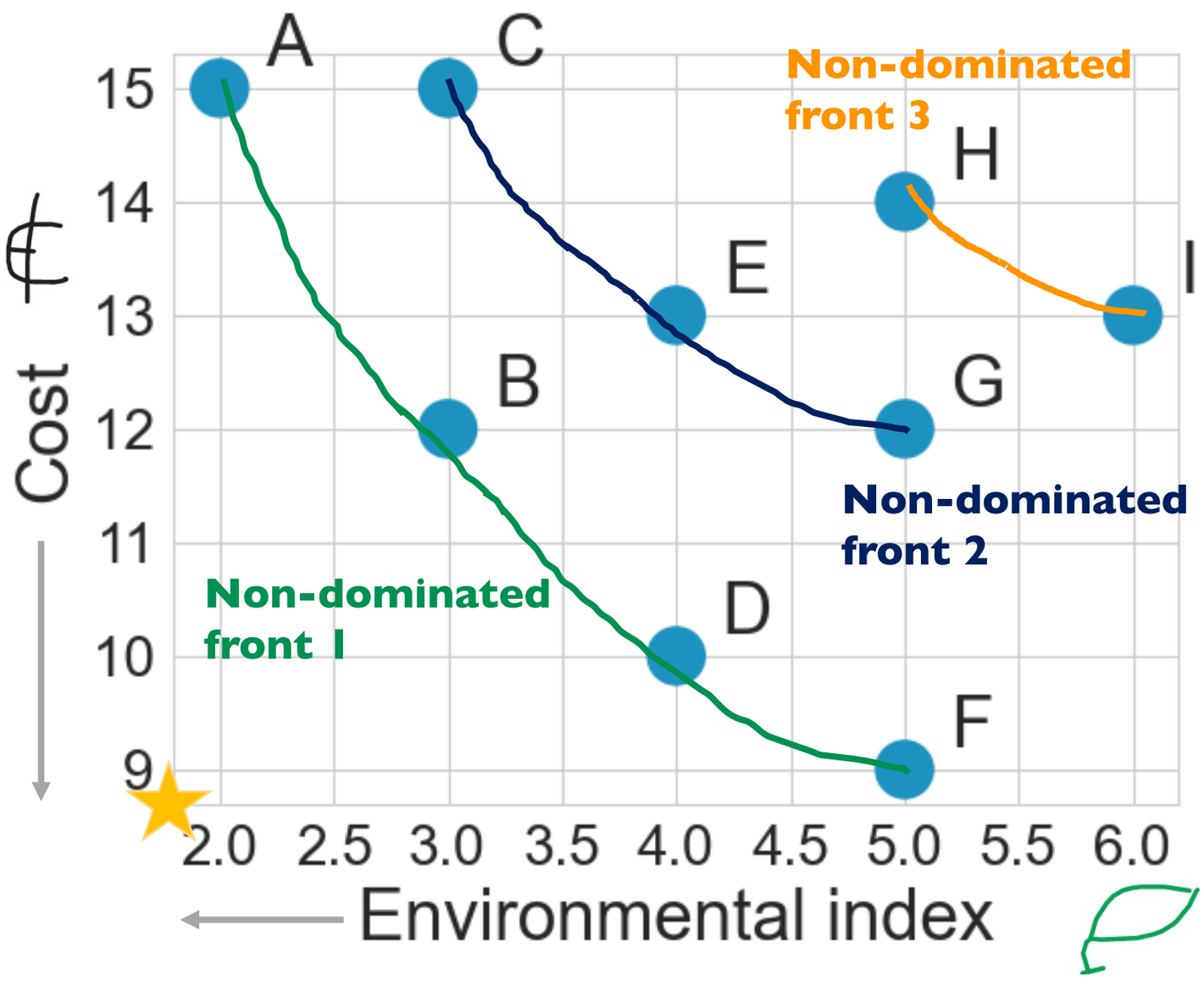


NSGA II Example: Pareto Fronts

Car type	Env index (min)	Cost (min)	Points dominated	No. points that dominate	Pareto rank
A	2	15K	C	0	1
B	3	12K	C,E,G,H,I	0	1
C	3	15K	-	0	2
D	4	10K	E,G,H,I	0	1
E	4	13K	H,I	0	2
F	5	9K	G,H,I	0	1
G	5	12K	H,I	0	2
H	5	14K	-	0	3
I	6	13K	-	0	3



NSGA II Example: Pareto Fronts



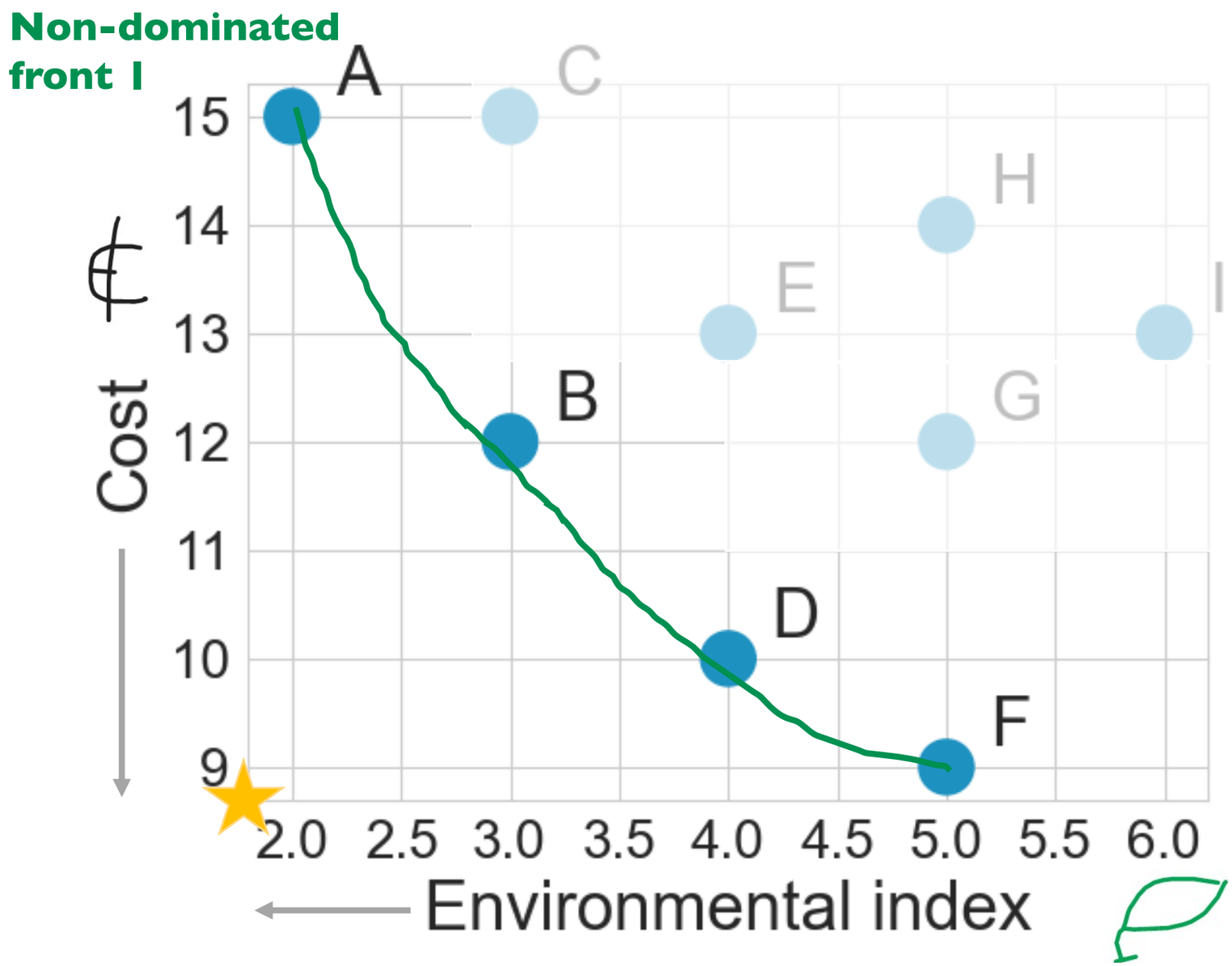
Fitness assignment:

-Non-dominated front 1 has the highest fitness

-Same front has the same fitness

-Lower non-dominated rank is preferred over others

NSGA II Example: Preserving Diversity



If we could only keep 3 solutions which ones should we keep?

NSGA II Example: Crowding Distance (CD)

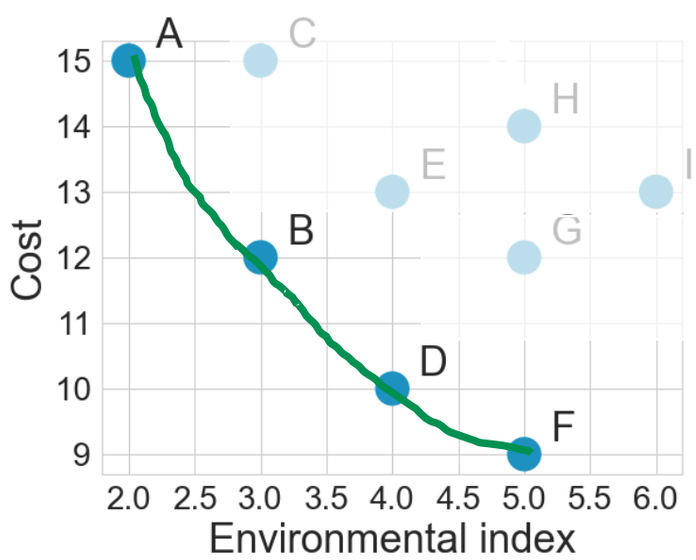
$$CD_{im} = \frac{f_m(x_{i+1}) - f_m(x_{i-1})}{f_m(x_{max}) - f_m(x_{min})}$$

$$CD_i = \sum_{m=1}^{M} CD_{im}$$

Car type	Env index (min)	Cost (min)	CD
A	2	15K	inf
B	3	12K	1.33
D	4	10K	1
F	5	9K	inf

Car type	Env index (min)	Cost (min)
A	2	15K
B	3	12K
C	3	15K
D	4	10K
E	4	13K
F	5	9K
G	5	12K
H	5	14K
I	6	13K

max	6	15
min	2	9
diff	4	6

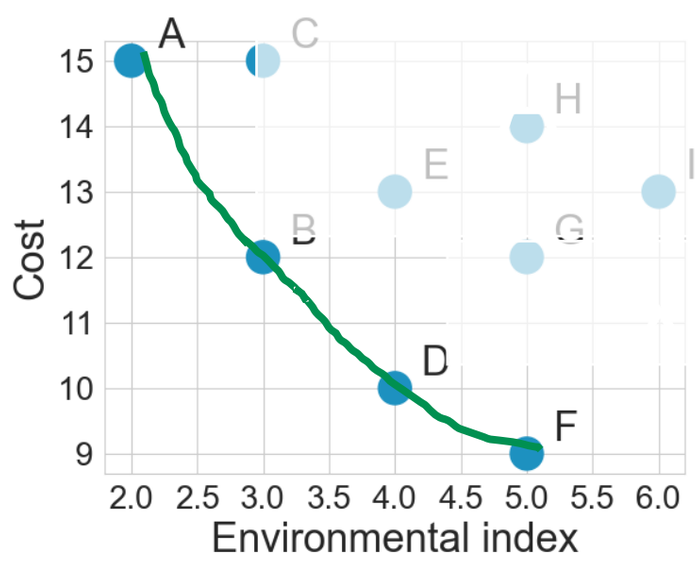


$$CD_B = \frac{4-2}{4} + \frac{15-10}{6} \approx 1.33$$

$$CD_D = \frac{5-3}{4} + \frac{12-9}{6} = 1$$

NSGA II Example: Preserving Diversity

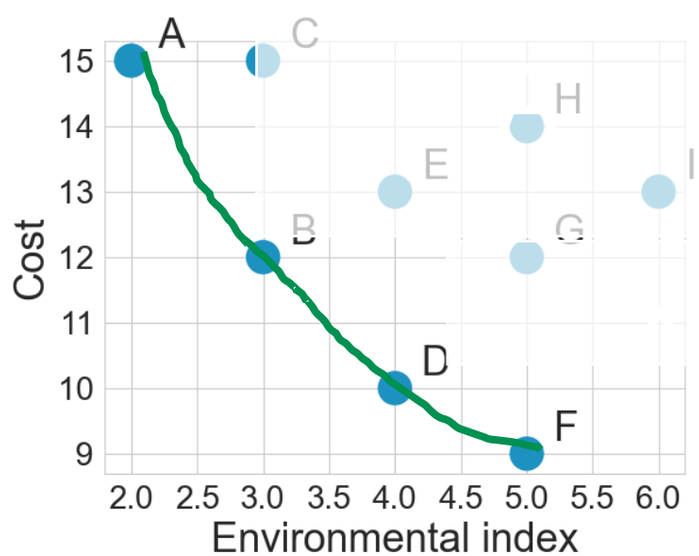
If you could only keep 3 solutions which point would you eliminate?



Options:	Car type	Env index (min)	Cost (min)	CD
A)	A	2	15K	inf
B)	B	3	12K	1.33
C)	D	4	10K	1
D)	F	5	9K	inf

NSGA II Example: Preserving Diversity

If you could only keep 3 solutions which point would you eliminate?



Options:	Car type	Env index (min)	Cost (min)	CD
A)	A	2	15K	inf
B)	B	3	12K	1.33
C)	**D**	**4**	**10K**	**1**
D)	F	5	9K	inf

A larger crowding distance is preferred to preserve diversity

NSGA II Example: Crowding Distance (CD)

$$CD_{im} = \frac{f_m(x_{i+1}) - f_m(x_{i-1})}{f_m(x_{max}) - f_m(x_{min})}$$

$$\boxed{CD_i = \sum_{m=1}^{M} CD_{im}}$$

!!!

Car type	Env index (min)	Cost (min)	CD
A	2	15K	inf
B	3	12K	1.33
D	4	10K	1
F	5	9K	inf

Car type	Env index (min)	Cost (min)
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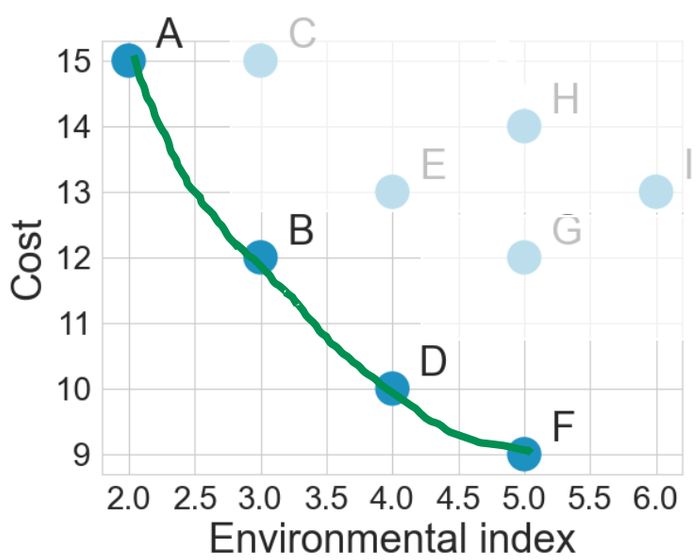
G	5	12K
H	5	14K
I	6	13K

max	6	15
min	2	9
diff	4	6



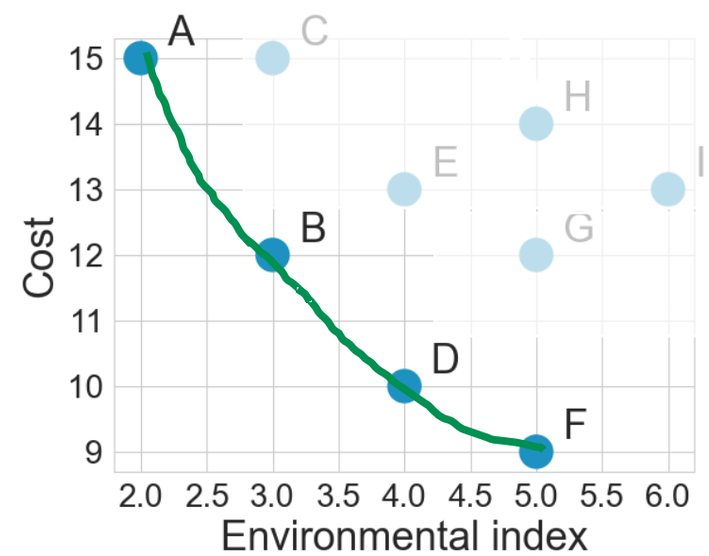
$$CD_B = \frac{4-2}{4} + \frac{15-10}{6} \approx 1.33$$

$$CD_D = \frac{5-3}{4} + \frac{12-9}{6} = 1$$

NSGA II Example: Crowding Distance (CD)

Sorted by env index (ascending)

Car type	Env index (min)	Cost (min)	CD
A	2	15K	inf
B	3	12K	1.33
D	4	10K	1
F	5	9K	inf



Sorted by env index (ascending)

Car type	Env index (min)	Cost (min)	CD
F	5	9K	inf
D	4	10K	1
B	3	12K	1.33
A	2	15K	inf

$$CD_B = \frac{4-2}{4} + \frac{15-10}{6} \approx 1.33$$

$$CD_D = \frac{5-3}{4} + \frac{12-9}{6} = 1$$

NSGA II Example: Crowding Distance (CD) – General Case

Input: $Q = \{x_1, x_2, \dots, x_l\}$ with the same rank

For each j, set $Q[j]_{distance} = 0$

For each objective f_i

$Q = sort(Q, f_i)$;

$Q[1]_{distance} = \infty$;

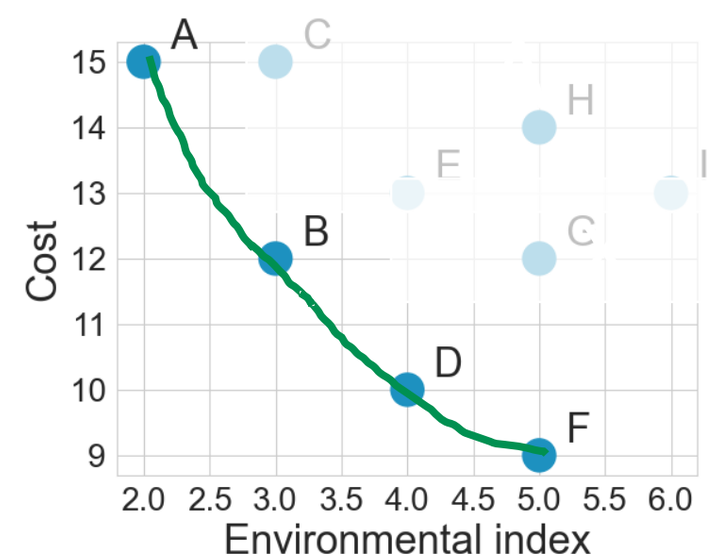
$Q[l]_{distance} = \infty$;

for $j = 2\ to\ l - 1$

$$Q[j]_{distance} += \frac{f_i(Q[j+1]) - f_i(Q[j-1])}{f_{i,max} - f_{i,min}}$$

end for

end for



Crowding distance: the larger the better

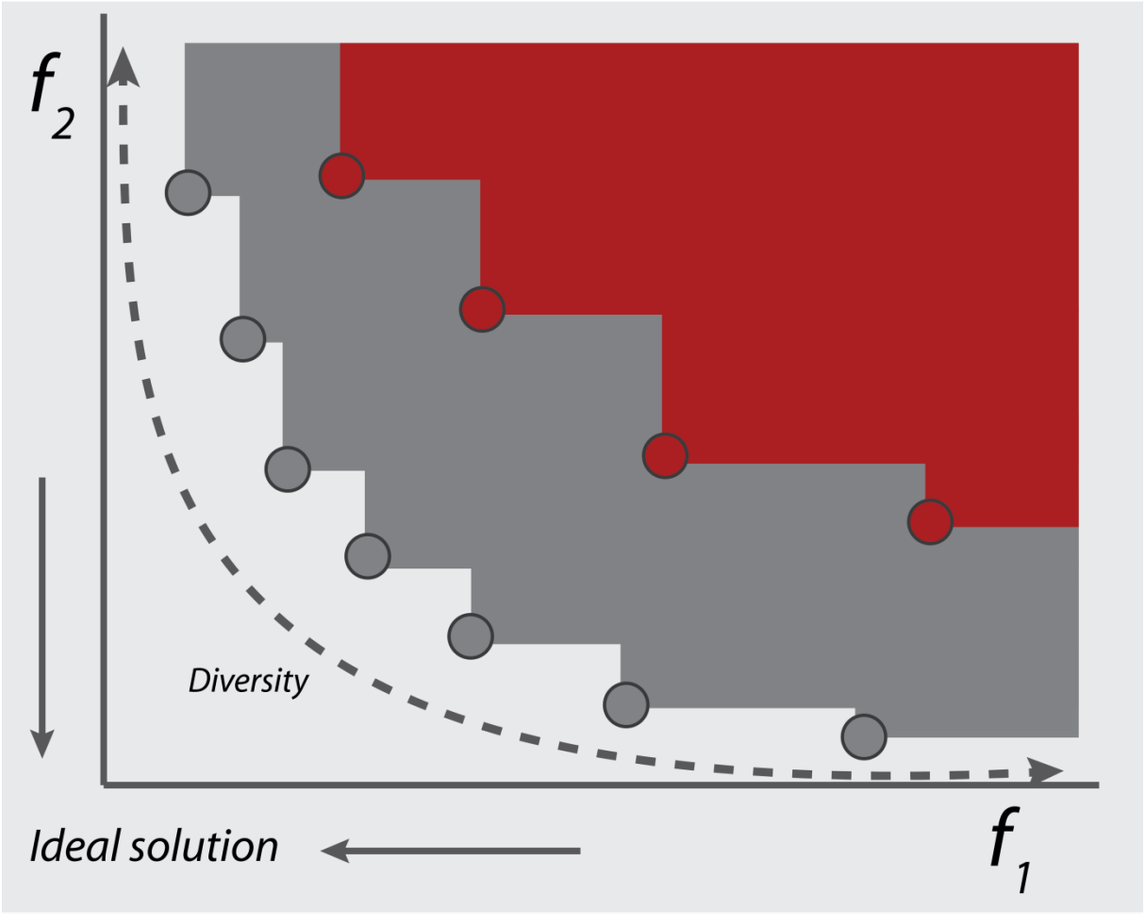
NSGA II: Fitness Strategies

Selection pressure mechanism:

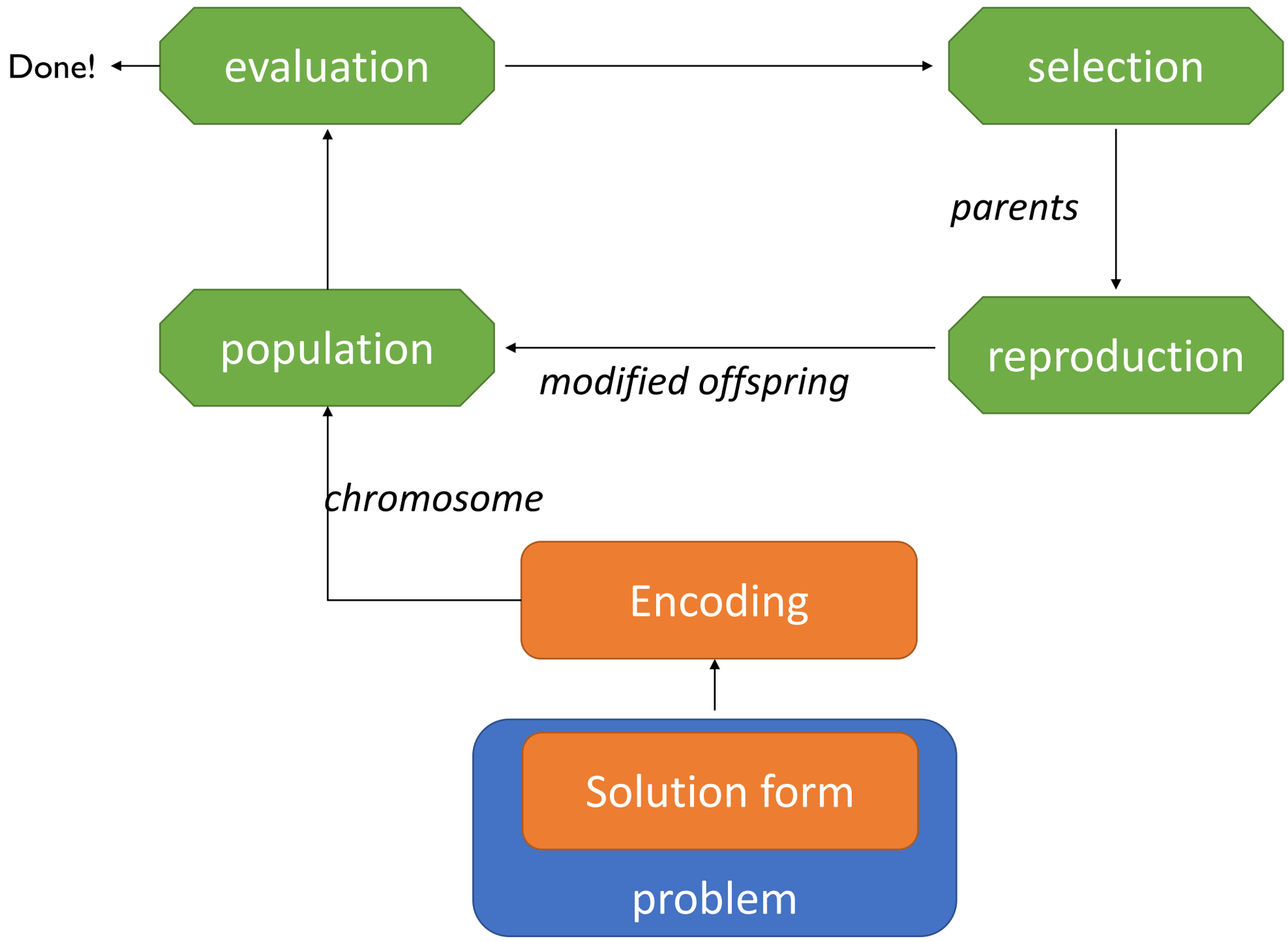
Non-dominated sorting

Diversity preservation:

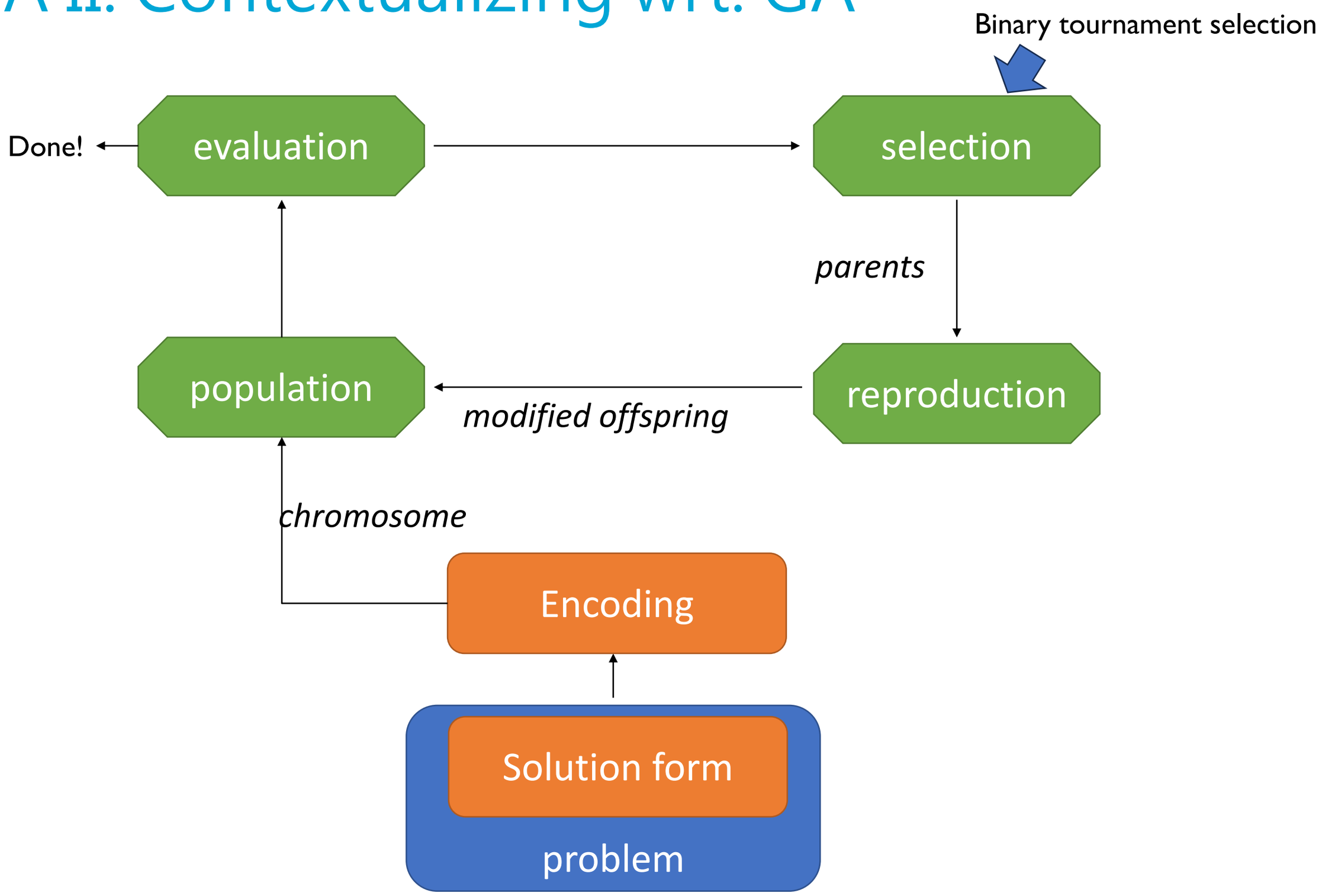
Crowding distance



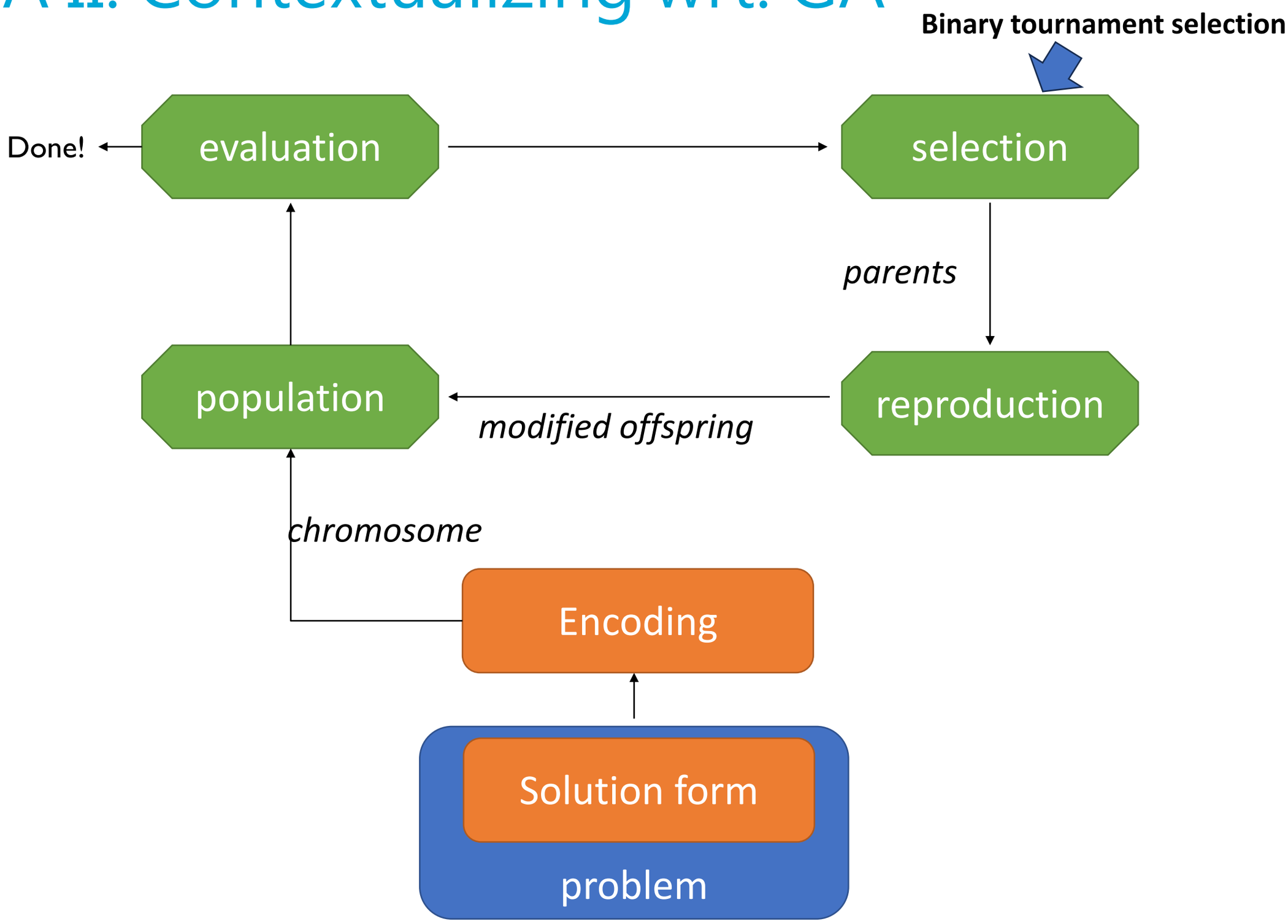
NSGA II: Contextualizing wrt. GA



NSGA II: Contextualizing wrt. GA



NSGA II: Contextualizing wrt. GA

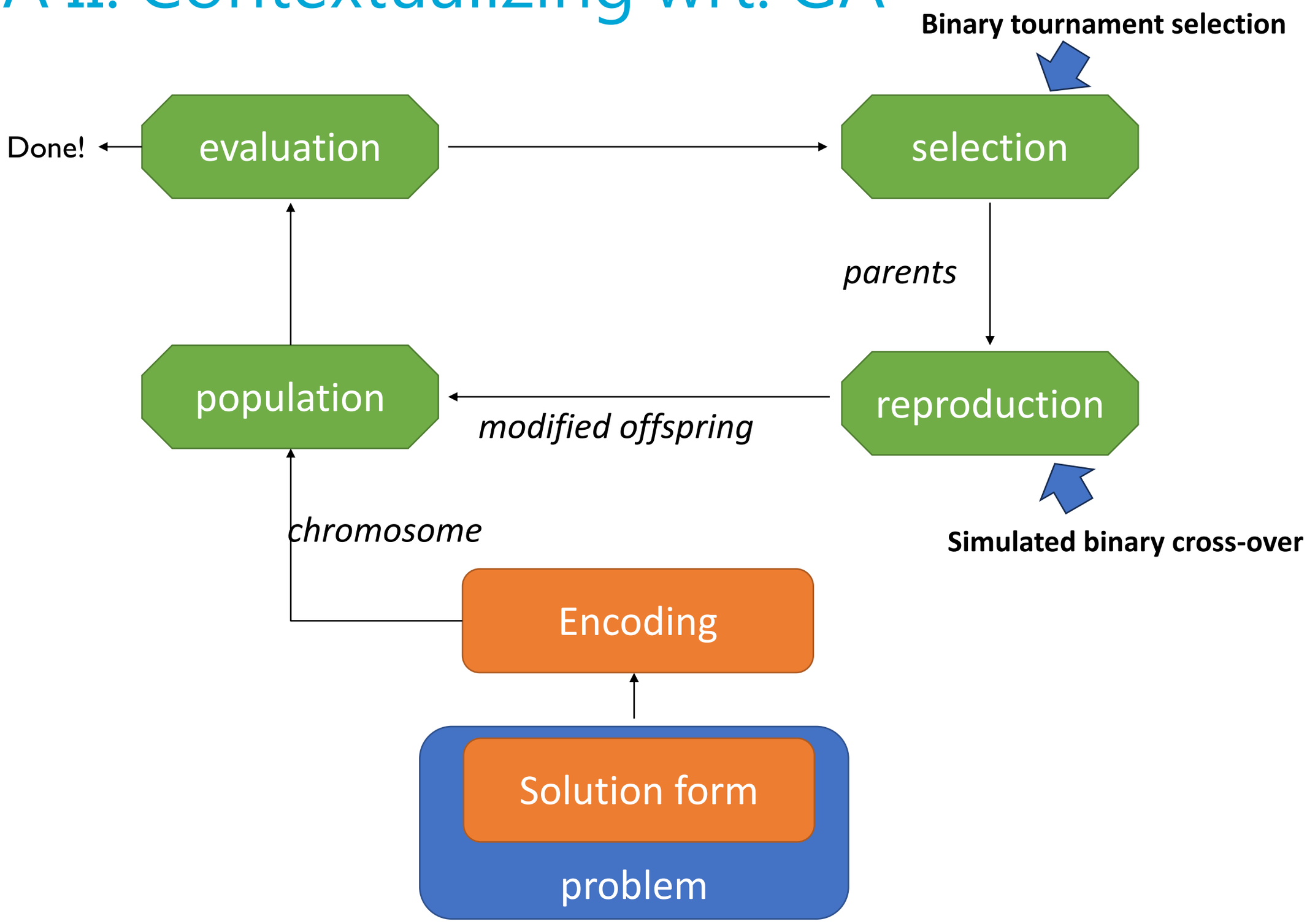


NSGA II: Contextualizing wrt. GA

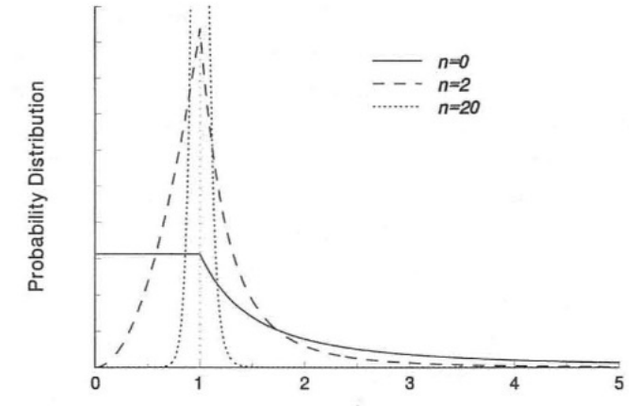
- Binary tournament selection
  - Pick two: have a tournament
  - How to have a tournament: compare rank (priority) then compare crowding distance
  - The winner becomes a parent, until the population is filled

- Scenario
  - **Candidate A:** A Rank 1 car. It has a great cost and environmental score, but there are 5 other cars with almost the exact same specs. Its **Crowding Distance is very low (e.g., 0.1)**.
  - **Candidate D:** A Rank 2 car. Its specs aren't quite as good as Candidate A. Its **Crowding Distance is very high (e.g., 2.5)**.
  - A wins!

NSGA II: Contextualizing wrt. GA



NSGA II: Contextualizing wrt. GA



- Simulated binary cross-over

$$c_1 = 0.5\left[(p_1 + p_2) - \beta_1'|p_2 - p_1|\right],$$
$$c_2 = 0.5\left[(p_1 + p_2) + \beta_2'|p_2 - p_1|\right].$$

$$\beta = \left|\frac{c_1 - c_2}{p_1 - p_2}\right|.$$

- What about mutation?
  - Polynomial mutation: uses a probability distribution to determine how much to change a value

$$p' = \begin{cases} p + \bar{\delta}_L(p - x_i^{(L)}), & \text{for } u \leq 0.5, \\ p + \bar{\delta}_R(x_i^{(U)} - p), & \text{for } u > 0.5. \end{cases} \quad (1)$$

Then, either of the two parameters ($\bar{\delta}_L$ or $\bar{\delta}_R$) is calculated, as follows:

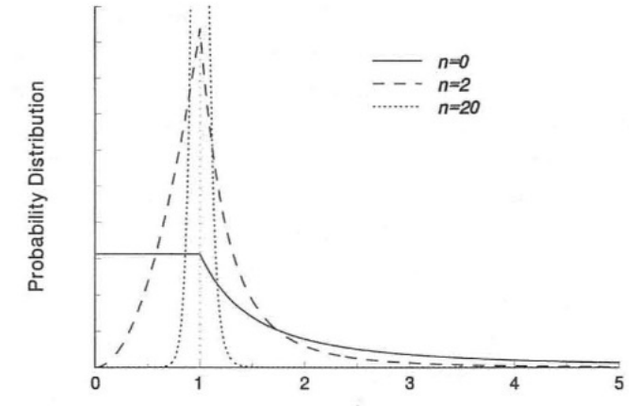
$$\bar{\delta}_L = (2u)^{1/(1+\eta_m)} - 1, \quad \text{for } u \leq 0.5, \quad (2)$$
$$\bar{\delta}_R = 1 - (2(1-u))^{1/(1+\eta_m)}, \quad \text{for } u > 0.5. \quad (3)$$

You will be not tested on this

https://content.wolfram.com/sites/13/2018/02/09-6-1.pdf

https://www.egr.msu.edu/~kdeb/papers/k2012016.pdf

NSGA II: Contextualizing wrt. GA



- Simulated binary cross-over

$$c_1 = 0.5\left[(p_1 + p_2) - \beta_1'|p_2 - p_1|\right],$$
$$c_2 = 0.5\left[(p_1 + p_2) + \beta_2'|p_2 - p_1|\right].$$

$$\beta = \left|\frac{c_1 - c_2}{p_1 - p_2}\right|.$$

- What about mutation?
  - Polynomial mutation: uses a probability distribution to determine how much to change a value

$$p' = \begin{cases} p + \bar{\delta}_L(p - x_i^{(L)}), & \text{for } u \le 0.5, \\ p + \bar{\delta}_R(x_i^{(U)} - p), & \text{for } u > 0.5. \end{cases} \quad (1)$$

Then, either of the two parameters ($\bar{\delta}_L$ or $\bar{\delta}_R$) is calculated, as follows:

$$\bar{\delta}_L = (2u)^{1/(1+\eta_m)} - 1, \quad \text{for } u \le 0.5, \quad (2)$$
$$\bar{\delta}_R = 1 - (2(1-u))^{1/(1+\eta_m)}, \quad \text{for } u > 0.5. \quad (3)$$

https://content.wolfram.com/sites/13/2018/02/09-6-1.pdf

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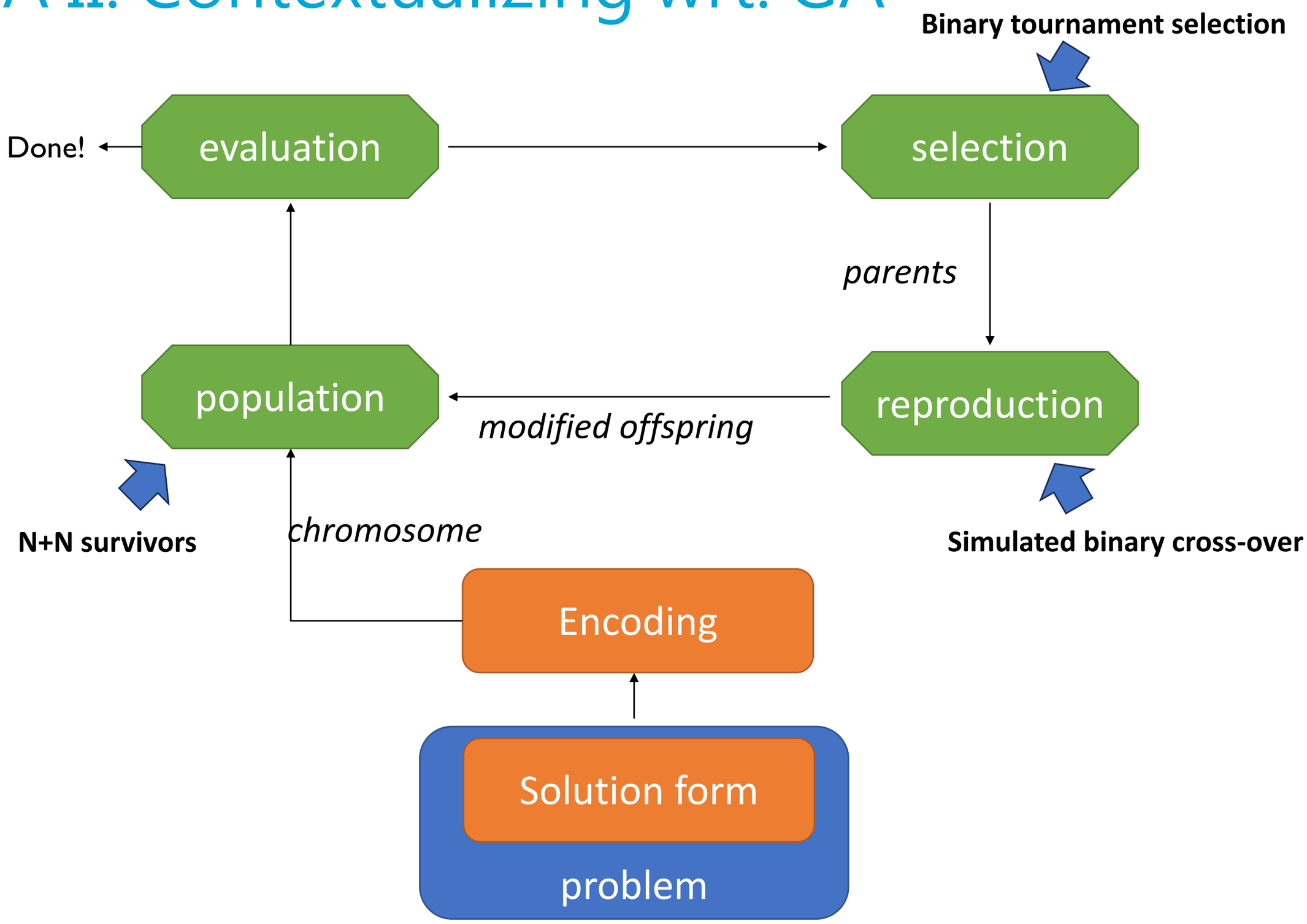
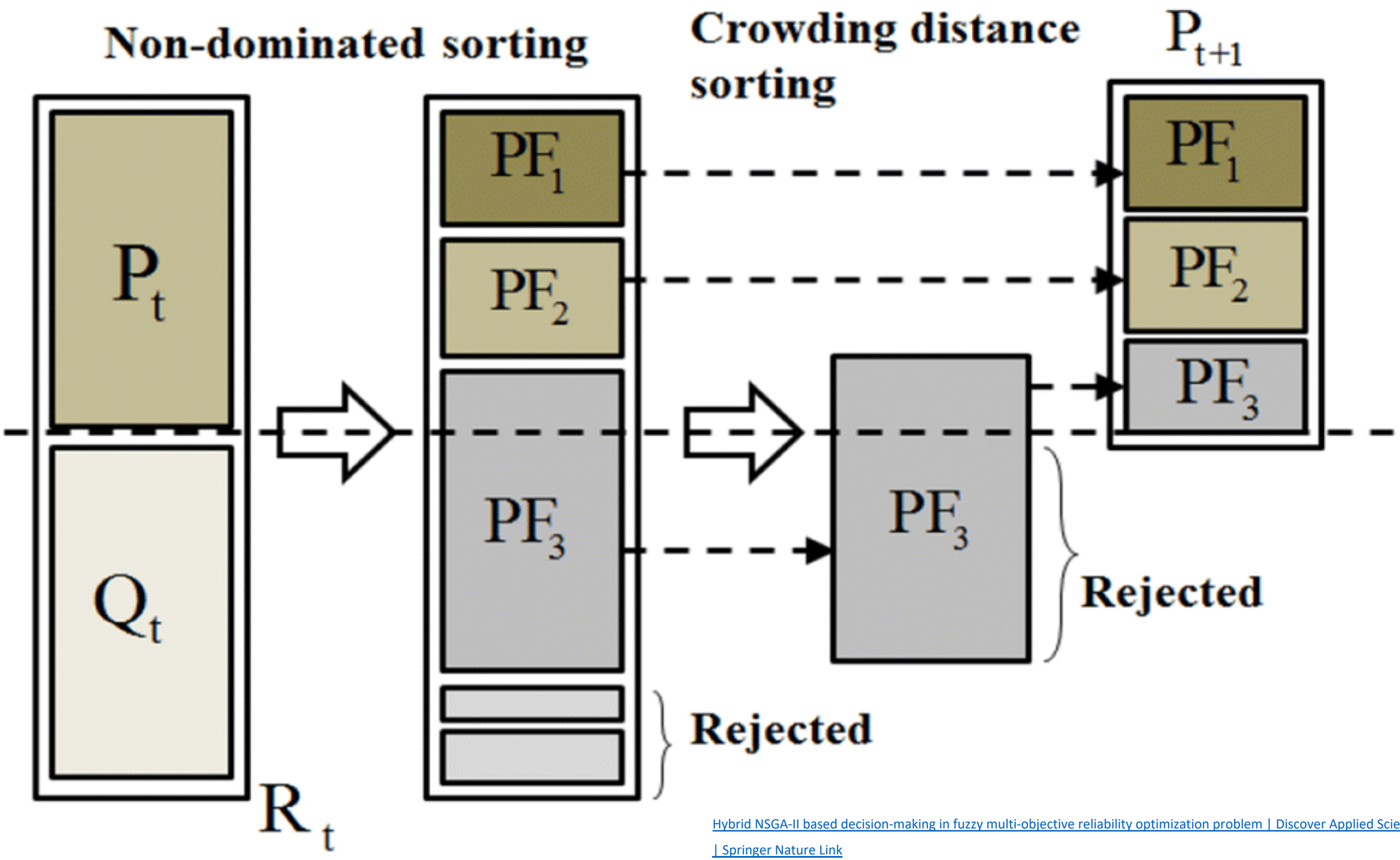
NSGA II: Contextualizing wrt. GA



NSGA II: Contextualizing wrt. GA



Hybrid NSGA-II based decision-making in fuzzy multi-objective reliability optimization problem | Discover Applied Sciences | Springer Nature Link

Multi-Objective Optimization: Wrap Up

- ✓ Multiobjective optimization problem consists of a problem with 2 or more objectives.
- ✓ Concept of Pareto optimality and non-dominance, allows to rank solutions in a multiobjective optimization problem
- ✓ Goals of multi-objective evolutionary optimization:
  - ✓ Preserve the best individuals (non-dominated sorting)
  - ✓ Preserve diversity (crowding distance)

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Summary

- Steps of Genetic Algorithm
- Applications of Genetic Algorithm
- Multi-Objective Evolutionary Optimization
- NSGA II Algorithm
- **Optional Reading from Chapter 7**: Evolutionary Strategies and Genetic Programming

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Slide references

- **Machine Learning — Evolutionary Algorithms**
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- Qian Chao:Heuristic Search and Evolutionary Algorithms, lec12 Evolutionary Algorithmsfor Multi-objective Optimization
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